ABSTRACT

This project behaves as a proof of concept, and explores the potential and viability of using computer vision and deep learning techniques for lip reading, specifically, the classification of words using purely visual data of the lip movements of an individual. It investigates the performance of various computer vision techniques and neural networks under varied environmental constraints and the feasibility of deploying them in real world solutions. The project makes use of a dataset specifically created for this study. It contained 2 second videos of 24 unique words, each spoken by 4 different speakers under a constrained environment. Our goal in this project was to study and demonstrate the techniques in computer vision and deep learning required to achieve visual speech recognition at the level of individual words. We tested with preprocessing techniques such as lip segmentation using 'dlib [26]', various architectures like 3DCNN, and Bidirectional LSTM to document the performance. While the end results of our study depict the feasibility of our endeavor, there were various impediments encountered along the way, such as overfitting, memory limitation and a limited dataset due to storage constraints. This work contributes to the field of visual speech recognition, and conveys the promise of CV and DL techniques in the advancement to real word solutions. Future work may investigate the use of a more extensive and varied dataset, refine models, and improve performance, potentially making way for solutions in the field of human-computer interactions (HCI), surveillance, and more.

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ABBREVIATIONS

VSR (visual speech recognition): A process of interpreting spoken words by analyzing visual information, such as lip movements, facial expressions, and gestures, typically using machine learning models.

GPU (Graphics Processing Unit): A specialized processor designed to accelerate rendering of images and parallel processing tasks, commonly used in machine learning and gaming.

VR (Virtual Reality): A simulated environment created using computer technology that immerses users in a fully artificial digital world.

AR (Augmented Reality): Technology that overlays digital content, such as images or information, onto the real world through devices like smartphones or AR glasses.

CNN (Convolutional Neural Network): A deep learning algorithm designed for

processing structured grid data like images, widely used in image recognition and computer

vision tasks.

RNN (Recurrent Neural Network): A type of neural network where connections between nodes form a directed graph along a temporal sequence, allowing it to process sequential data like time series or text.

3DCNN (3D Convolutional Neural Network): An extension of CNNs that processes three-dimensional data, often used for analyzing videos or volumetric medical images.

LSTM (Long Short-Term Memory): A type of RNN designed to learn long-term dependencies in sequential data by managing memory through gates.

BiLSTM (Bidirectional Long Short-Term Memory): An LSTM network that processes data in both forward and backward directions, improving context understanding in sequential tasks.

HCI (Human-Computer Interaction): The study and practice of designing user-friendly interfaces that allow effective interaction between humans and computers.

ReLU [28] (Rectified Linear Unit): An activation function in neural networks that outputs the input if it's positive, otherwise zero, helping to introduce non-linearity.

TCN (Temporal Convolutional Network): A neural network architecture designed for sequential data processing, using dilated convolutions to capture long-range dependencies.

HMM (Hidden Markov Model): A statistical model that represents systems with hidden states and observable outputs, commonly used in time-series analysis and speech recognition.

SVM (Support Vector Machine): A supervised machine learning algorithm that separates data points using a hyperplane to perform classification or regression tasks.

CHAPTER 1

INTRODUCTION

Lip-reading, also known as visual speech recognition (VSR), is the process of recognizing speech by visually interpreting the movements of the lips, face, and other facial features. It has long been a vital tool for individuals with hearing impairments, helping them to understand spoken language in environments where sound is either unavailable or unreliable. However, the potential applications of lip-reading extend far beyond accessibility. In recent years, there has been growing interest in applying lip-reading techniques to a variety of domains, including security and surveillance, human-computer interaction, and virtual reality.

The rise of deep learning and computer vision has brought new opportunities to the field of lip-reading. Previously, lip-reading relied heavily on manual feature extraction and traditional machine learning techniques, which limited its scalability and accuracy. However, the advent of **neural networks**, particularly **convolutional neural networks (CNNs)** and **recurrent neural networks (RNNs)**, has dramatically improved the performance of visual speech recognition systems. These advanced models are capable of automatically learning relevant features from video data, capturing the complex dynamics of lip movements in real-time. This shift has made it feasible to classify spoken words from video data, making the task of lip-reading more accurate and applicable to diverse real-world scenarios.

One of the key challenges in lip-reading technology lies in dealing with the multimodal nature of speech. Human speech involves not only vocal cord vibrations but also lip movements, nasal control, etc. which must all be accounted for to ensure high accuracy. Modern machine learning algorithms, particularly those using **spatio-temporal features**, are well-suited to handle this complexity, as they can process both spatial (lip shape) and temporal (lip movement over time) information in parallel.

Moreover, large datasets are crucial for training deep learning models to effectively recognize speech patterns from video. With the increasing computational power of modern GPUs, the field of visual speech recognition is poised to make significant advancements. The ongoing research and development in this area are not only pushing the boundaries of what is possible in lip-reading but also laying the foundation for integrating these systems into everyday applications, such as real-time communication systems, privacy-preserving security surveillance, and more accessible technological interfaces for people with hearing loss.

1.1. Problem Statement

The challenge of accurately interpreting spoken words based solely on visual information, such as lip movements, presents a significant hurdle in the field of **visual speech recognition (VSR)**. Traditional speech recognition systems rely heavily on **audio data**, making them ineffective in scenarios where sound is unavailable, such as noisy environments or silent communication. Despite advancements in machine learning and computer vision, lip-reading systems still face key obstacles, including the **variability of lip movements** across different speakers, **limited datasets**, and the **multimodal nature of speech** (which involves not only lip movements but also voice intonations and facial expressions).

To address these challenges, this project explores the feasibility of applying deep learning models to classify spoken words based on video data. These models are well-suited for capturing the spatio-temporal dynamics of lip movements over time. By focusing on facial data preprocessing, including lip-region detection using dlib [26] and CNNs for feature extraction, the project aims to demonstrate that lip-reading is a viable solution to overcome the limitations of traditional speech recognition systems.

While this project serves as a proof of concept, the goal is to assess the effectiveness of deep learning architectures for visual speech recognition and explore their potential applications in fields such as accessibility, security, and human-computer interaction. The project's results will help inform future advancements in lip-reading technology and its integration into real- world applications. [31]

1.2. Motivation for Project Work

1. **Real-World Applications**:
    - **Accessibility**: Visual speech recognition has the potential to significantly aid people with hearing impairments, offering an alternative way of understanding speech in environments where audio cannot be relied upon (e.g., noisy spaces or silent communication in video calls).
    - Security and Surveillance: Lip-reading is valuable in security contexts where audio recording is prohibited or unclear, allowing for non-invasive interpretation of conversations from video footage in surveillance situations. [32]
    - **Human-Computer Interaction**: Lip-reading can enhance human-computer interaction, enabling intuitive, gesture-based communication, especially in silent or noisy environments (e.g., public spaces) where conventional speech recognition may fail.
    - **Virtual Reality (VR) and Augmented Reality (AR)**: In VR/AR systems, lip-reading could enable immersive communication without the need for microphones, supporting more natural and seamless interactions within virtual environments.

2. **Challenges with Current Systems**:
   - Traditional speech recognition systems rely primarily on **audio data**, which can be unreliable under certain conditions (e.g., noisy environments, low-quality recordings, or when speech is muted). Visual speech recognition, leveraging lip movements and facial expressions, offers a promising complementary solution.
   - While **deep learning** and **computer vision** have proven effective for a range of speech-related tasks, their application to lip-reading remains underexplored. There is a pressing need for research to validate the feasibility and refine the performance of such technologies in real-world scenarios.

3. **Advancements in Machine Learning**:
   - The development of **deep learning** models, particularly **neural networks**, has made it possible to automatically **extract relevant features** from video data, eliminating the need for manual feature engineering. This represents a significant improvement over previous methods, which were limited in scalability and performance.
   - With the availability of improved **computational resources** (such as **GPUs**), there is an opportunity to push the boundaries of visual speech recognition. This project aims to explore how deep learning models can be applied to lip-reading tasks, such as **word classification** from lip movements.

4. **Proof of Concept and Feasibility**:
   - Although advances have been made, there remains a gap in demonstrating the **practical feasibility** of applying deep learning models to real-world lip-reading tasks. This work also aims to investigates the performance of various neural network architectures in this domain for classifying words from video data.
   - The project serves as a proof of concept, demonstrating the feasibility of visual speech recognition while addressing the challenges of applying deep learning to lip-reading. The results aim to provide valuable insights into the viability of lip-reading for real-world applications, including accessibility, security, and human-computer interaction.

1.3. Scope of Project

This project aims to explore and implement machine learning techniques for **visual speech recognition (VSR)**, specifically focusing on **lip-reading** from video data. The scope includes the development and evaluation of neural network models to classify spoken words based on **lip movements**, with a particular focus on utilizing deep learning methods for **feature extraction**, **data preprocessing**, and **model training**.

Key aspects of the project include:

1. **Dataset Creation and Data Collection: [33]**
    - The project utilizes a **custom dataset** consisting of **24 words** spoken by **4 different speakers** under controlled conditions. The dataset is designed to focus on **lip movements** specific to the selected words, enabling the study of word-level classification.
2. **Preprocessing and Feature Extraction**:
    - The preprocessing pipeline involves detecting and isolating the lip region using tools like dlib [26] and preparing the video frames for training. The project also explores the use of 3DCNN and BiLSTM for capturing both the spatial and temporal features of the lip movements.
3. **Model Training and Evaluation**:
    - The focus of the project is on training **neural networks**, such as **3DCNN** and **BiLSTM**, to classify words based on **lip movement patterns**. Various models will be tested and compared to assess their effectiveness in visual speech recognition tasks.
4. **Challenges and Limitations**:
    - The project explores challenges such as **overfitting**, **memory constraints**, and **data sparsity**—issues that are common when working with small, controlled datasets. It aims to identify strategies to improve model performance under these constraints.
5. **Future Work and Potential Applications**:
    - This project lays the groundwork for future work in **lip-reading** and **visual speech recognition** by demonstrating the feasibility of the approach with a small dataset. The results have the potential to contribute to **accessibility** (for individuals with hearing impairments), **security** (through surveillance applications), and **human-computer interaction (HCI)**, where visual data can supplement or replace traditional audio-based recognition systems.

1.4. Algorithms Used

In this project, various algorithms and techniques were employed to address the challenges of lip-reading word classification, such as high-dimensional input data, temporal dependencies, and variability across speakers and sessions. These methods were chosen to enhance model accuracy, efficiency, and generalization.

1. **3D Convolutional Neural Networks (3D CNN [28]s):**
    - Purpose: Extract spatiotemporal features from lip-region video frames. [34]
    - Implementation: The 3D CNN [28] processes sequences of frames, capturing spatial

features (e.g., lip contours) and their temporal dynamics (e.g., lip movement patterns).

- **Advantage:** Effective for modeling the sequential nature of lip movements while preserving spatial details.

2. **Bidirectional Long Short-Term Memory Networks (BiLSTM):**
   - **Purpose:** Model temporal dependencies and contextual relationships in lip movement sequences.
   - Implementation: BiLSTM layers were used after the 3D CNN [28] to process frame- level features in both forward and backward temporal directions.
   - **Advantage:** Captures long-term dependencies in lip movements, improving word classification accuracy.
3. **ConvLSTM2D [13]:**
   - **Purpose:** Handle spatiotemporal data by combining the strengths of convolutional and recurrent layers.
   - Implementation: ConvLSTM2D [13] was used to process video sequences, where each frame is treated as a 2D spatial structure, and temporal connections are established across frames. [35]
   - **Advantage:** Simultaneously learns spatial and temporal patterns, making it well-suited for tasks involving sequential visual data.
4. **Attention Mechanisms:**
   - **Purpose:** Highlight the most relevant spatiotemporal features in the input sequence.
   - **Implementation:** Attention layers were integrated with BiLSTM to focus on key regions of interest in the frame sequences.
   - **Advantage:** Enhanced feature discrimination by prioritizing important temporal and spatial information.
5. **Dropout:**
   - **Purpose:** Mitigate overfitting and improve model generalization.
   - **Implementation:** Dropout layers were applied at various stages of the model, randomly deactivating neurons during training.
   - **Advantage:** Prevents reliance on specific neurons, ensuring robust performance across different speakers and sessions.
6. **Batch Normalization:**
   - **Purpose:** Stabilize training by normalizing intermediate activations.
   - **Implementation:** Batch normalization was applied in the CNN layers to maintain consistent activation distributions and accelerate convergence.
   - **Advantage:** Improves training stability and allows for higher learning rates.
7. **L1 and L2 Regularization:**
   - **Purpose:** Penalize complex models to reduce overfitting and encourage simpler solutions.
   - **Implementation:** L1 and L2 regularization were applied to the model's weights, adding constraints to prevent over-reliance on specific features.
   - **Advantage:** Improved generalization by reducing model complexity and avoiding overfitting.

8. **ReLU [28] Activation Function:**
   - **Purpose:** Introduce non-linearity in the model to capture complex patterns in data.
   - Implementation: ReLU [28] was used in convolutional and dense layers to activate neurons based on positive values only.

- **Advantage:** Efficient computation and reduced vanishing gradient issues during training.

9. **Softmax Activation Function:**
   - **Purpose:** Convert raw model outputs into probabilities for multi-class classification.
   - **Implementation:** Softmax was applied in the final layer of the network to predict the likelihood of each word class.
   - **Advantage:** Ensures outputs are interpretable as probabilities, aiding in classification tasks.
10. **Adam Optimizer:**
    - **Purpose:** Optimize model parameters by minimizing the loss function.
    - Implementation: The Adam optimizer [28] was employed due to its adaptive learning rate capabilities and robustness to noisy gradients.
    - **Advantage:** Faster convergence and reliable performance across diverse training conditions.
11. **Mixed Precision Training:**
    - **Purpose:** Optimize computational resources while maintaining model accuracy.
    - **Implementation:** Mixed precision training enabled the use of float16 operations for faster computation, while preserving critical calculations in float32.
    - **Advantage:** Reduced training time and efficient GPU memory utilization.

1.5. Research Gaps:

1. **Robustness in Noisy Environments**: While lipreading technology shows promise in noisy conditions, many studies still struggle with achieving high accuracy when external noise interferes.
2. Generalization Across Diverse Populations: Many existing models are trained on specific datasets that may not represent diverse demographics. There is a need for research that addresses speaker variability [8], including different accents, ages, and ethnicities, to ensure models generalize well across various populations.
3. **Temporal Feature Extraction**: Although some studies have utilized LSTMs and TCNs for temporal modeling, there remains a gap in effectively capturing long-range dependencies in lip movements.
4. **Real-Time Processing**: Many current systems are not optimized for real-time applications. Research into more efficient algorithms that can process visual data quickly without sacrificing accuracy would be beneficial, particularly for applications in assistive technologies.
5. **Dataset Limitations**: Although several datasets have been established (e.g., GRID, LRS), there is a lack of large-scale, publicly available datasets that include diverse speaking styles and conditions.
6. Attention Mechanisms: While attention-based models have been explored, there is still room for innovation in how attention mechanism [20]s are implemented to focus on relevant features during lipreading tasks.
7. **Feature Extraction Techniques**: Many studies rely heavily on CNNs for feature

extraction but do not explore hybrid approaches that might combine traditional feature extraction methods with deep learning techniques.

1.6. Objectives of Present Work:

This project serves as a **proof of concept** to validate the feasibility of developing an accurate visual-based lip-reading model and establish a flexible testing pipeline for exploring and comparing neural network architectures. The objectives are as follows:

1. **Develop a Visual-Based Lip-Reading Model:** Create a robust system that relies on visual input to classify spoken words based on lip movements.
2. **Achieve High Accuracy in Constrained Data Environments:** Attain an accuracy of over 80% in scenarios where data is consistent and follows well-defined constraints.
3. **Test Performance on Data with Limited Constraints:** Achieve an accuracy of over 60% in the final stage of testing, which involves data with greater variability and fewer constraints.
4. **Establish a Flexible Testing Pipeline:** Design and implement a pipeline that enables testing of various neural network architectures under controlled and reproducible conditions. This pipeline ensures seamless integration, evaluation, and comparison of models, promoting adaptability and iterative improvements to the system.

By achieving these objectives, the project validates the core methodologies for visual-based lip reading while providing a scalable framework for further research and development in this domain.

1.7. Methodology

This chapter outlines the methodology used in this project, including the data collection process, preprocessing steps, model selection, and training pipeline [28]. The goal is to create a pipeline that can be easily replicated by others for visual speech recognition using our custom dataset.

1. **Utilization of Custom Lip-Reading Dataset:**
   - The custom dataset consists of 24 words spoken by 4 speakers, with each word spoken 10 times per speaker in separate sessions. Each session contains 10 videos of a single speaker saying the same word. These videos are stored in session-based folders, and the labels are extracted from the file names during loading, avoiding the need for an external metadata file. [36]
   - The data collection process is automated, ensuring consistency and reproducibility. This automated pipeline allows others to replicate the data collection for their own tests and to build their custom datasets for similar tasks.
2. **Data Preparation:**

- **Preprocessing:**
  - Videos are cropped around the lip region using dlib [26] for facial landmark detection, resized to 64x64 or 128x128 pixels, and converted to grayscale to reduce computational complexity.
  - The number of frames per video is standardized during preprocessing, either

by padding or truncating videos to a consistent frame length.

- **Feature Extraction:**
  - The feature extraction process depends on the selected model architecture. For **3DCNNs**, features are extracted from the video frames, while for **BiLSTM**, extracted frame sequences are passed through the LSTM layers to capture sequential dependencies.
- **Data Split:**
  - A **stratified split** is used to ensure that each split contains a balanced distribution of classes (words) across training and validation sets.

3. **Training Pipeline:**
   - **Feature Extraction:**
     - Models are trained using appropriate loss functions and optimizers to minimize errors during training. Hyperparameters such as batch size and the number of epochs are tuned based on validation performance.
   - **Evaluation:**
     - Model performance is assessed using standard metrics such as accuracy, precision, recall, and F1-score. Confusion matrices are also analyzed to identify common misclassifications.

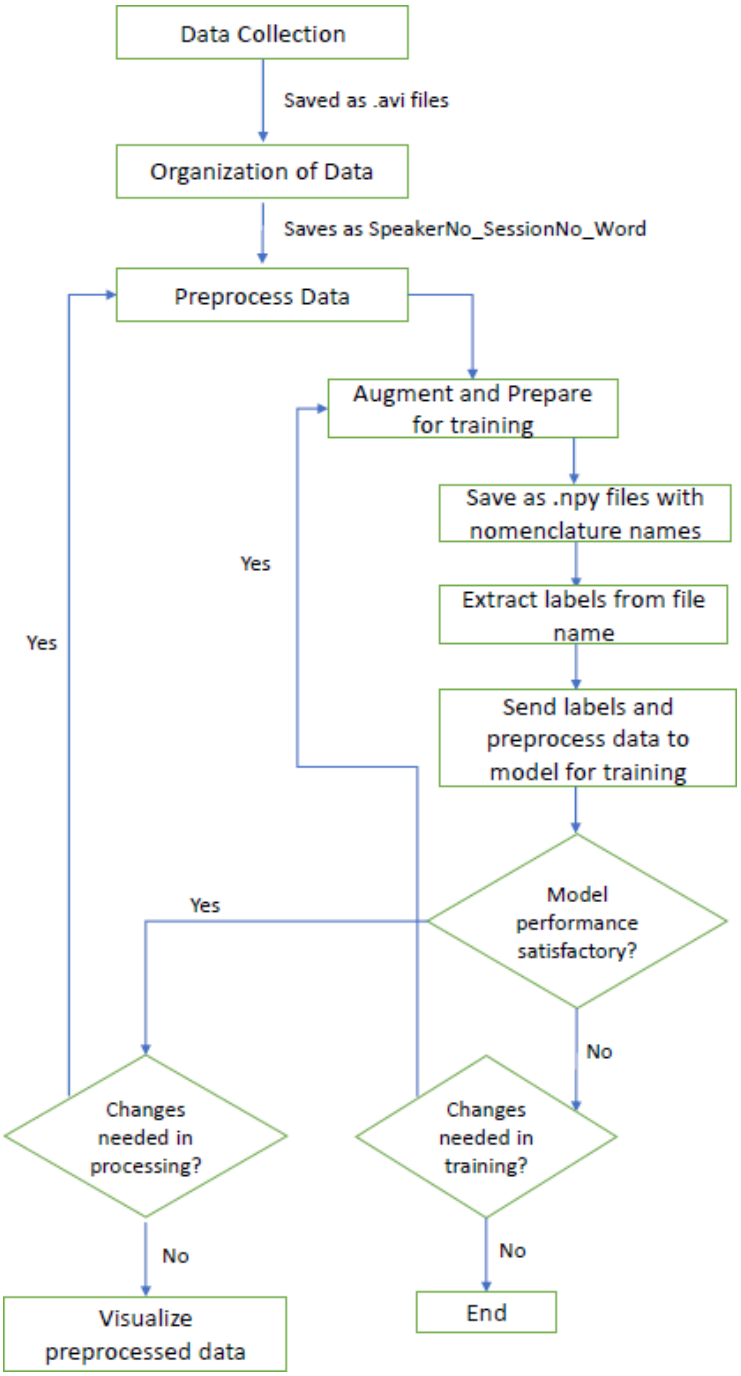


Figure 1: Flowchart of project

4. **Reproducibility:**
   - The automated data collection and preprocessing pipeline, along with the detailed model setup, ensures that the experiments can be easily replicated by others. This methodology allows for testing and comparing various models for visual speech recognition, using custom datasets and a consistent, reproducible process.

CHAPTER 2

LITERATURE SURVEY

SR No.	Authors	Year	Title	Methodology	Key Findings	Dataset Used
1	M. Hao et al.	2020	A Survey of Research on Lipreading Technology	Review of traditional methods (HMMs, PCA) and deep learning methods (CNNs, LSTMs)	Discusses evolution from traditional techniques to modern approaches.	Various datasets discussed, including LRW and LRS.
2	Amit Garg et al.	2021	Lip Reading Using CNN and LSTM	CNN for spatial feature extraction; LSTM for sequence modeling	Improved accuracy by combining CNN for spatial features and LSTM for temporal modeling.	GRID dataset
3	Atharva Karekar et al.	2021	Lip Reading Using Deep Learning	CNNs and RNNs for feature extraction and classification	Demonstrated effectiveness of deep learning models for visual speech recognition tasks.	Not specified
4	Yannis M. Assael et al.	2017	LipNet: End-to-End Sentence-Level Lipreading	Spatial-temporal convolution networks; CTC loss function	Achieved high accuracy in sentence-level lipreading with novel architecture using CTC loss function.	LRS dataset
5	Grace Tilton	2018	Computer Vision Lip Reading	Various computer vision techniques (specific methods not detailed)	Explored various approaches to enhance lipreading capabilities.	Not specified

SR No.	Authors	Year	Title	Methodology	Key Findings	Dataset Used
6	Themos Stafylakis & Georgios Tzimiropoulos	2018	Combining Residual Networks with LSTMs for Lipreading	Residual Networks combined with LSTMs	Improved recognition rates by combining residual networks with LSTMs.	GRID dataset
7	Gaoyan Zhang & Yuanyao Lu	2022	Research on a Lip-Reading Algorithm Based on Efficient-GhostNet	Efficient-GhostNet architecture; specific optimization techniques	Introduced an efficient model reducing complexity while maintaining accuracy in lipreading tasks.	Not specified
8	Hongyang Huang et al.	2022	A Novel Machine Lip Reading Model	Novel deep learning model (specific architecture not detailed)	Proposed a new architecture improving recognition performance with multiple feature extraction techniques.	Not specified
9	Gutiérrez, Abiel and Zoe - Alanah Robert	2018	Lip Reading Word Classification	CNNs for feature extraction; SVM for classification	Focused on classifying words based on lip movements using deep learning techniques.	Not specified
10	Weicong Chen et al.	2020	DualLip: A System for Joint Lip Reading and Generation	Dual-stream network combining lip reading and generation tasks	Developed a system that integrates lip reading with speech generation, enhancing	Not specified

SR No.	Authors	Year	Title	Methodology	Key Findings	Dataset Used
					contextual understanding.	
11	N. Deshmukh et al.	2021	Vision based Lip Reading System using Deep Learning	CNNs and LSTMs for feature extraction and classification	Proposed a vision-based system utilizing deep learning for improved lip reading accuracy.	Not specified
12	K. Neeraja et al.	2021	Deep Learning based Lip Movement Technique for Mute	CNNs and RNNs for lip movement synthesis	Developed a technique to synthesize lip movements for individuals who are mute using deep learning models.	Not specified
13	Ümit Atila et al.	2022	Turkish Lip-Reading Using Bi-LSTM and Deep Learning Models	Bi-LSTM networks for temporal modeling; CNNs for feature extraction	Achieved promising results in Turkish lip-reading tasks using Bi-LSTM architectures.	Not specified
14	Stillittano & Caplier	2008	Inner Lip Segmentation by Combining Active Contours and Parametric Models	Active contours combined with parametric models	Presented a method for segmenting inner lips effectively using contour-based techniques.	Not specified
15	Swami & Jayasimha	2021	Lip Reading Recognition	HMMs and neural networks	Focused on improving recognition rates through advanced	Not specified

SR No.	Authors	Year	Title	Methodology	Key Findings	Dataset Used
					modeling techniques.	
16	Oscar Koller et al.	2017	Re-Sign: Re-Aligned End-to-End Sequence Modeling with Deep Recurrent CNN-HMM	Deep Recurrent CNN-HMM model for sequence alignment	Proposed an end-to-end model that improves alignment in sequence modeling tasks.	Not specified
17	Zeng et al.	2020	Deep Triplet Neural Networks with Cluster-CCA for Audio-Visual Cross-Modal Retrieval	Deep triplet neural networks; Cluster-CCA	Enhanced retrieval performance by learning a shared embedding space for audio-visual data.	AVE dataset
18	Brais Martinez	2020	Lipreading Using Temporal Convolutional Networks	Temporal Convolutional Networks (TCNs)	Demonstrated effectiveness of TCNs in capturing temporal dependencies in lip movements.	GRID dataset
19	Priyanka P. Kapkar & S. D. Bharkad	2019	Lip Feature Extraction and Movement Recognition Methods	Various feature extraction methods; classification algorithms	Reviewed multiple methods for extracting lip features and recognizing movements effectively.	Various datasets discussed, including GRID and LRS.

SR No.	Authors	Year	Title	Methodology	Key Findings	Dataset Used
20	Lu, Y., & Li, H.	2019	Automatic Lip-Reading System Based on Deep Convolutional Neural Network and Attention-Based Long Short-Term Memory	Deep CNNs combined with Attention-based LSTM	Achieved improved recognition rates by utilizing attention mechanisms to focus on relevant lip features.	GRID dataset

Table 1

Technologies Used in Lipreading Research Papers

1. **Convolutional Neural Networks (CNNs)**
   - **Purpose**: CNNs are used for feature extraction from lip images due to their ability to automatically learn spatial hierarchies of features. They excel at recognizing patterns in visual data, making them suitable for lipreading tasks.
   - **Papers**: Noda et al. (2014), Gutiérrez et al. (2018), Lu & Li (2019).

2. **Long Short-Term Memory Networks (LSTMs) [37]**
   - **Purpose**: LSTMs are employed for sequence modeling, capturing temporal dependencies in lip movements over time. This is crucial for understanding the dynamics of speech as it unfolds.
   - **Papers**: Amit Garg et al. (2021), Yannis M. Assael et al. (2017), Lu & Li (2019).

3. **Deep Learning Techniques**
   - **Purpose**: General deep learning frameworks are used to automate feature extraction and improve classification accuracy by learning from large datasets without manual intervention.
   - **Papers**: Atharva Karekar et al. (2021), Hongyang Huang et al. (2022).

4. **Temporal Convolutional Networks [18] (TCNs)**
   - **Purpose**: TCNs are utilized for capturing temporal dependencies in sequences, similar to LSTMs but with a different architectural approach that can be more efficient in certain contexts.
   - Papers: Brais Martinez (2020). [38]

5. **Hidden Markov Models [15] (HMMs)**
   - **Purpose**: HMMs have been traditionally used for modeling sequences and are effective in tasks where temporal patterns are important, such as speech recognition.
   - **Papers**: Oscar Koller et al. (2017), Swami & Jayasimha (2021).

6. **Support Vector Machines (SVM)**
   - **Purpose**: SVMs are used as classifiers for distinguishing between different lip movement patterns based on extracted features.
   - **Papers**: Gutiérrez et al. (2018).

7. **Attention Mechanisms**
   - **Purpose**: Attention mechanisms help models focus on relevant parts of the input data, improving performance by emphasizing important features during the learning process.
   - **Papers**: Lu & Li (2019).

8. **Dual-Stream Networks**
   - **Purpose**: These networks combine different modalities, such as audio and visual data, to enhance performance in tasks like joint lip reading and speech generation.
   - **Papers**: Weicong Chen et al. (2020).

9. **Active Contours [14] and Parametric Models**
   - **Purpose**: These techniques are used for segmenting lips accurately from video frames, which is critical for effective feature extraction.
   - **Papers**: Stillittano & Caplier (2008).

10. **Color-Based Feature Extraction Methods**
    - **Purpose**: Various color-based methods are employed to enhance lip segmentation by differentiating lip colors from surrounding skin tones.
    - **Papers**: Priyanka P. Kapkar & S. D. Bharkad (2019).

CHAPTER 3

PRELIMINARIES

This section provides an overview of the key concepts, tools, and technologies used in this project. Understanding these fundamentals is crucial for grasping the methodology and how the visual speech recognition system is developed. The section covers the main techniques used for lip-reading, including deep learning architectures, video preprocessing [28], and evaluation metrics.

3.1. Deep Learning Architecture for VSR

Several deep learning models are commonly used for tasks like visual speech recognition. For this project, two key model types are explored:

1. **Convolutional Neural Networks (CNNs):**

   CNNs are used to capture spatial features from each frame of the video. These features correspond to the visual cues in the lip region, such as the shape and movement of the lips. CNNs are effective at extracting hierarchical features from images, which makes them suitable for the initial processing of video frames.

2. **Recurrent Neural Networks (RNNs):**

   RNNs, particularly **Bidirectional Long Short-Term Memory (BiLSTM)** networks, are designed to capture temporal dependencies in sequential data. In the case of VSR, RNNs process the extracted spatial features from each frame to understand the sequence of lip movements and associate them with specific words. BiLSTMs, in particular, can capture both past and future dependencies, making them effective for speech recognition tasks.

3.2. Preprocessing Techniques for Video Data:

Before training a model, the raw video data needs to be preprocessed to ensure it is in a format that is suitable for input to deep learning models. Key preprocessing steps include:

1. **Facial Landmark Detection:**

   Using tools like dlib [26], facial landmarks are detected to crop the video frames around the lip region. This reduces the amount of unnecessary background information and focuses on the area most relevant for lip-reading.

2. **Frame Resizing and Normalization:**

   Video frames are resized to a consistent dimension (e.g., 128x128 pixels) to ensure that the model receives input in a uniform size. Additionally, frames are converted to grayscale to reduce computational complexity and are normalized to have pixel values within a specific range (e.g., 0 to 1).

3. **Frame Consistency: [39]**

   Videos are processed to ensure that they have a consistent number of frames, either by padding shorter videos or truncating longer ones. This step ensures that the neural network receives input with consistent time steps across all training samples.

3.3. Data Augmentation:

While not used in this specific project, data augmentation [8] techniques are commonly employed in visual speech recognition tasks to artificially increase the size of the dataset. Common augmentation [8] methods include:

1. **Random Scaling and Rotation:**

   Slightly altering the position or scale of the face can help the model generalize better to different viewpoints.

2. **Temporal Warping:**

   Modifying the speed of the video, stretching, or compressing the time axis, can simulate different speaking speeds.

Data augmentation [8] can be particularly useful when working with small datasets or when attempting to improve model robustness to various real-world conditions.

3.4. Evaluation Metrics:

To evaluate the performance of visual speech recognition models, several key metrics are typically used:

1. **Accuracy:**
   The percentage of correctly predicted words or phonemes relative to the total number of predictions.

2. **Precision, Recall, and F1-Score:**

   These metrics are particularly useful when dealing with imbalanced classes or when the focus is on minimizing false positives or false negatives.

3. **Confusion Matrix:**

   A confusion matrix helps visualize the performance of the classification model by showing the number of correct and incorrect predictions for each class, making it easier to identify common misclassifications.

4. **Loss Functions:**

   During model training, loss functions (e.g., categorical cross-entropy [28]) quantify the error between predicted and actual labels, guiding the optimization process.

3.5. Hardware and Software Tools:

The following tools and technologies are used for implementing and testing the visual speech recognition system:

1. **TensorFlow/PyTorch:**
   These deep learning frameworks are used for building, training, and evaluating the models. They offer high-level APIs for implementing complex architectures like CNNs and LSTMs.
2. **dlib [26]:**
   A library used for facial landmark detection, enabling the extraction of the lip region for preprocessing. [40]
3. **OpenCV:**
   A popular computer vision library used for video processing tasks, such as reading video files, frame extraction, and basic image transformations.
4. **CUDA (NVIDIA):**

   CUDA is used for GPU acceleration during model training to speed up the process, especially when working with large video datasets.

CHAPTER 4

IMPLEMENTED WORK

This chapter outlines the work implemented in this project, covering the data collection process, preprocessing steps, model selection, and training pipeline [28]. The goal is to build a reproducible pipeline for visual speech recognition (lip-reading) that can be easily replicated by others using custom datasets. This section covers the specifics of the dataset creation, preprocessing techniques, model architectures, and evaluation of the trained models.

4.1. Creation and Preparation of Custom Dataset:

The creation and preparation of the custom dataset were executed using three primary scripts: **Video Collector**, **Data Organizer**, and **Preprocessor**. These scripts were designed to automate the collection, organization, and preprocessing of video data to ensure efficient model training.

4.1.1. Video Collector

Libraries: OpenCV for video capture and processing.

Functionality:

- **Captures Video**: The system captures short video clips of speakers articulating target words. Each video clip represents one instance of the word being spoken.
- **Even Distribution of Words**: The system ensures an even distribution of words across instances by adhering to predefined recording criteria, such as speaker diversity and word repetition.
- **Storage Format**: All video clips are saved in a consistent format (e.g., .avi or .mp4), with clear naming conventions to allow for easy processing and labeling.

Implementation Details:

- The video capture process was automated to minimize user intervention and ensure consistency. Each video clip was automatically saved with a filename indicating the speaker, session, and word (e.g., Speaker1_Session1_apple).
- The raw video data served as the foundational input for the subsequent preprocessing steps.

Outcome:

A repository of uniformly captured raw video samples, with metadata linked to the corresponding words, speakers, and sessions.

4.1.2. Data Organizer

Libraries: Python OS module for file organization.

Functionality:

- **File Naming**: Videos were renamed to follow a consistent structure: SpeakerNo_SessionNo_Word. This structure was designed to ensure clarity and consistency across sessions and speakers.
- **Folder Organization**: The videos were organized into speaker- and session-specific subdirectories, allowing for easy navigation and access during training.
- **Label Extraction**: The intuitive naming conventions facilitated the extraction of labels for each video (e.g., "apple", "banana") during training.

Implementation Details:

- Custom Python scripts automated the renaming and structuring of video files, reducing manual effort and the possibility of errors in data organization.
- The organizational structure allowed seamless integration with the preprocessing pipeline, ensuring the videos were ready for subsequent steps.

4.1.3. Preprocessing and Visualization

Libraries: CuPy for GPU acceleration, TensorFlow for optimized memory usage, and Dlib for facial landmark detection.

Functionality:

- **Lip Region Extraction**: Dlib's facial landmark detection was used to extract the lip region from each frame. This step focused on the most relevant visual information for lip-reading. [41]
- **Frame Resizing**: All frames were resized to a 128x128 resolution to ensure consistency across all videos.
- **Grayscale Conversion**: Each frame was converted to grayscale, retaining the intensity information while reducing computational complexity.
- **Video Length Standardization**: Videos were padded or trimmed to ensure a consistent length of 60 frames, providing a uniform input for neural networks.
- **Augmentation**: Data augmentation [8] techniques (brightness adjustment, contrast adjustment, noise addition, rotation, and scaling) were applied to improve model robustness and generalization.

- **Visualization**: Preprocessed videos were visualized to ensure the quality and consistency of the output, allowing for any issues to be detected and corrected early.

Implementation Details:

- Preprocessing was batch-processed for efficiency, utilizing GPU acceleration with CuPy to reduce processing time.
- The augmentation techniques helped simulate a variety of real-world conditions (lighting changes, head movement, etc.), enhancing the diversity of training data.
- Quality checks were implemented after each preprocessing batch to ensure the processed videos met the expected standards.

Outcome:

- A comprehensive and standardized dataset, with augmented variants, was created, ensuring it was ready for training. [42]

4.2. Preprocessing Techniques:

The preprocessing phase was crucial to preparing the data for optimal model training. Key methods included:

4.2.1 Frame Resizing:

- **Resolution**: Video frames were resized to 128x128 pixels, utilizing interpolation techniques that preserved the quality of the lip region.
- **Consistency**: This standardization ensured that the inputs to the network had uniform dimensions, which is essential for training.

4.2.2 Grayscale Conversion:

- **Efficiency**: Grayscale conversion simplified the data without losing critical visual features, such as the intensity variations in lip movement.
- **Reduced Complexity**: By eliminating the need to process color information, the network could focus on the more relevant spatial features.

4.2.3 Frame Padding or Trimming:

- **Uniform Length**: Videos were standardized to 60 frames, ensuring that all sequences had a consistent time dimension. Shorter videos were padded, and longer ones were trimmed.
- **Temporal Consistency**: This step ensured that the network could process sequential information in a stable manner.

4.2.4 Data Normalization:

- Normalization: Pixel intensity values were scaled to the range [0, 1] by dividing by 255. This helped improve the convergence speed during training and maintained numerical stability. [43]

4.2.5 Augmentation Techniques:

Several augmentation techniques were employed:

- **Brightness Adjustment**: Improved model robustness to different lighting conditions.
- **Contrast Adjustment**: Enhanced performance when the contrast in the video varied.
- **Gaussian Noise Addition**: Simulated noise, increasing the model's ability to handle noisy input.
- **Rotation and Scaling**: Simulated head movements and changes in the distance between the camera and the speaker, enhancing generalization.

4.2.6 Lip Region Extraction:

- **Dlib's Facial Landmark Detection**: Used to extract the lip region from each video frame, ensuring that the model focused only on the relevant visual information.
- **Improved Accuracy**: This step was vital for improving the accuracy of the system by eliminating unnecessary features (e.g., background) and focusing only on the lip movements.

Outcome:

Preprocessing techniques ensured a high-quality, standardized, and augmented dataset ready for deep learning models.

4.3. Model Training:

The training phase involved the implementation of various neural network architectures and optimization strategies to classify words based on lip movements. We tested several models and optimized the pipeline for performance.

4.3.1 Model Architectures:

This section details the neural network architectures implemented to address the challenges of lip-reading word classification, emphasizing the extraction of spatiotemporal features and sequential patterns.

1. 3D Convolutional Neural Network (3D CNN)

The 3D CNN is a natural extension of 2D CNNs, designed to process spatiotemporal data such as video frames.

Input Layer:

- The model takes preprocessed 3D lip-region frames as input, formatted as (time x height x width x channels). Each input represents a sequence of grayscale frames capturing lip movements over time.

3D Convolutional Layers:

- These layers apply 3D kernels (filters) to extract both spatial and temporal features simultaneously.
- Each kernel scans the spatial dimensions (height, width) across consecutive frames (temporal dimension), capturing motion and changes in lip positions.

- **3D Max Pooling Layers**:
  - Downsamples the feature maps to reduce computational complexity and prevent overfitting while retaining important features.
  - Pooling operates over a 3D window, ensuring both spatial and temporal information is preserved.
- **Fully Connected Layers**:
  - After the convolutional layers, the extracted spatiotemporal features are flattened and passed through dense layers for classification.
  - The final softmax layer outputs probabilities for each word class.

Key Strengths:

The ability to simultaneously capture motion dynamics (temporal) and lip shape variations (spatial) makes 3D CNNs highly effective for video data.

2. Attention Mechanism

The attention mechanism dynamically focuses on the most relevant frames or features in the input sequence, mimicking human cognitive focus.

- **Weight Calculation**:
  - Attention scores are computed for each frame or feature in the sequence. These scores represent the relative importance of each frame for the task.
  - Softmax is applied to normalize these scores into attention weights.
- **Feature Prioritization**:
  - The attention weights are used to compute a weighted sum of features, giving higher importance to critical frames while downplaying less relevant ones. [44]
- **Enhanced Interpretability**:
  - The mechanism improves model interpretability by allowing visualization of which parts of the input sequence were most significant for predictions.
  - In the context of lip reading, attention helps focus on frames with significant lip movements.

3. Long Short-Term Memory (LSTM):

LSTMs are a type of recurrent neural network (RNN) designed to model long- term dependencies in sequential data.

- **Cell States and Gates**: The

  LSTM cell consists of:

  - **Forget Gate**: Decides which information to discard from the previous cell state.
  - **Input Gate**: Selectively updates the cell state with new information.
  - **Output Gate**: Controls how much of the cell state is passed to the output.

These gates enable the model to retain relevant information over long sequences while forgetting irrelevant parts.

- **Sequential Data Handling**:
  - Lip movements often depend on prior frames, and LSTMs excel at capturing these dependencies.
  - By managing information flow through gates, LSTMs address vanishing gradient issues typical in vanilla RNNs.
- **Key Strengths**:

LSTMs are effective for handling variable-length sequences and long-term temporal dependencies in the data.

4. Bidirectional Long Short-Term Memory (BiLSTM):

BiLSTMs extend LSTMs by incorporating a second layer that processes the input sequence in reverse.

- **Bidirectional Processing**:
  - The forward pass processes the sequence from start to end, while the backward pass processes it from end to start.
  - Outputs from both directions are concatenated to capture contextual information from both past and future frames.

- **Enhanced Sequential Understanding**:
  - BiLSTMs enable the model to consider future frames when analyzing the current frame, leading to a deeper understanding of temporal patterns.
  - This is particularly useful in lip reading, where the meaning of a frame can depend on both preceding and succeeding movements.

Key Strengths:

By combining information from both directions, BiLSTMs improve

recognition accuracy for complex temporal patterns.

5. Convolutional Long Short-Term Memory (ConvLSTM2D):

ConvLSTM2D integrates convolutional operations into the LSTM framework to process spatiotemporal data.

- **Mechanism**:
  - Instead of using fully connected layers within the LSTM gates, ConvLSTM2D employs convolutional layers.
  - This allows the network to retain spatial structure while learning temporal dependencies. [45]
- **Gate Operations**:
  - Similar to standard LSTMs, ConvLSTM2D uses input, forget, and output gates to regulate information flow.
  - However, these gates operate on feature maps instead of flattened vectors, preserving spatial correlations.
- **Spatiotemporal Modeling**:
  - ConvLSTM2D processes each frame as a 2D feature map and considers its temporal relationship with previous frames.
  - This makes it particularly effective for tasks requiring simultaneous spatial and temporal feature extraction, such as lip movement recognition.

Key Strengths:

ConvLSTM2D excels at capturing complex spatiotemporal patterns, making it a robust choice for video-based tasks.

Outcome:
Five neural network architectures—3D CNN, Attention Mechanism, LSTM, BiLSTM, and ConvLSTM2D—were successfully implemented, each contributing unique strengths to the task of spatiotemporal feature extraction and word classification. These models provided a comprehensive exploration of different approaches to lip reading, enabling a detailed comparison of their capabilities.

4.3.2 Optimization, Loss Function, and Regularization Techniques

This section outlines the optimization strategies, loss functions, and regularization techniques applied to enhance model training, improve convergence, and mitigate overfitting.

1. Adam Optimizer:

The **Adam (Adaptive Moment Estimation)** optimizer is a popular choice for training deep learning models due to its efficiency and adaptive nature.

- **Mechanism**:
  - Adam combines the benefits of two other optimization techniques: [46]
    - **Momentum**: Uses an exponentially decaying average of past gradients to speed up convergence.
    - **RMSProp**: Scales learning rates based on the average of recent squared gradients for each weight.
  - It maintains two moving averages:
    - **Mean of gradients (m):** Helps adjust the direction of updates.
    - **Variance of gradients (v):** Controls the step size adaptively.
    - **Advantages**:

Adaptive learning rates for each parameter, reducing the need for manual tuning.

Robust performance on noisy gradients and sparse datasets.

Ensures faster and more stable convergence.

2. Sparse Categorical Crossentropy:

This loss function was chosen for the multi-class classification task, where the target labels were integers representing word classes.

- **Mechanism**:

The function calculates the cross-entropy loss between the predicted probability distribution and the true label.

In the sparse variant, true labels are provided as integers, which simplifies computation by avoiding one-hot encoding.

- **Formula**:

$$\text{Loss} = -\sum_{i} y_i \log(\hat{y}_i)$$

  - y_i: True class label (as an integer).
  - $\hat{y}_i$: Predicted probability for the correct class.

- **Advantages**:

  Efficient for multi-class classification tasks with a large number of classes.

  Reduces memory usage and computational complexity compared to standard categorical crossentropy.

3. Batch Normalization:

Batch normalization is a technique to standardize activations within a layer during training.

- **Mechanism**:

  For each mini-batch, the activations are normalized to have zero mean and unit variance:

$$\hat{x} = \frac{x - \mu}{\sqrt{\sigma^2 + \epsilon}}$$

  - x: Activation values.
  - μ, σ^2: Mean and variance of the current batch.
  - ϵ: Small constant to prevent division by zero.

  Scale and shift parameters are learned to allow the network to adjust the normalization as needed.

- **Benefits**:
  - Stabilizes learning by controlling internal covariate shift.
  - Speeds up convergence by allowing higher learning rates.
  - Regularizes the model, reducing overfitting when combined with dropout.

4. Dropout:

Dropout is a regularization technique designed to reduce overfitting by preventing the network from relying too heavily on specific neurons.

- **Mechanism**:

During training, a fraction of neurons is randomly deactivated (set to zero) in each layer.

This forces the network to learn redundant representations, enhancing generalization.

- **Implementation**:

A dropout rate (p) determines the probability of deactivating neurons. Typical values range from 0.2 to 0.5.

- **Advantages**:
  - Reduces overfitting by encouraging diversity in learned features.
  - Ensures the network remains robust during inference, as all neurons are used with scaled weights.

5. Mixed Precision Training:

Mixed precision training combines **float16** (half-precision) and **float32**

(single- precision) arithmetic to optimize performance on GPUs.

- **Mechanism**:
  - Computations are performed in float16 format, reducing memory usage and increasing throughput.
  - Key operations (e.g., loss scaling) are executed in float32 to maintain numerical stability.
- **Benefits**:
  - Reduces GPU memory consumption, enabling larger batch sizes.
  - Accelerates training by leveraging the Tensor Cores on modern GPUs.
  - Maintains accuracy comparable to full float32 training.
- **Implementation**:
  - Mixed precision was enabled using TensorFlow's tf.keras.mixed_precision module.
  - Loss scaling was applied to prevent underflow during backpropagation.

- **Outcome:**

By combining these techniques, the models achieved the following: [47]

- **Improved Performance**: Faster convergence and higher accuracy due to adaptive optimizations and effective loss computation.
- **Reduced Overfitting**: Techniques like dropout and batch normalization helped generalize the models to unseen data.
- **Enhanced Efficiency**: Mixed precision training significantly optimized GPU memory usage and training speed.

4.4 Attempted Models:

In addition to the models mentioned above, we experimented with four main neural network architectures:

- **3D CNN**: A standard convolutional network extended into the 3D domain to capture spatiotemporal features.
- **Attention-based 3D CNN**: Incorporated an attention mechanism to highlight important frames and features for classification.
- **ConvLSTM2D**: Combined convolutional layers with LSTM to model both spatial and temporal features simultaneously.
- **3D CNN + BiLSTM**: Integrated a 3D CNN for feature extraction followed by BiLSTM layers to capture temporal dependencies from both directions.

These networks allowed for comprehensive experimentation and comparison of their respective strengths in handling lip movements and classification tasks. [48]

4.5 Data Pipeline Integration:

The data pipeline integrated preprocessing, augmentation, and training workflows to enable seamless model development. This section describes the stages involved in preparing and feeding data into the model.

- **Outcome**:
    - A streamlined data pipeline was developed, ensuring efficient model training and evaluation.

4.6 Model Performance and Evaluation:

The trained models were evaluated based on their accuracy in word classification, using various metrics to gauge robustness and generalization.

- **Outcome**:
- Model performance was thoroughly evaluated, with results showing strong accuracy and generalization across the dataset.

CHAPTER 5

RESULT & ANALYSIS

5.1 Model Architectures and Training Configurations

To evaluate and compare the performance of different model architectures, the following four architectures were trained under identical, constrained conditions using 64x64 preprocessed data without augmentations:

1. **3D Convolutional Neural Network (3D CNN):**

- **Key Features:** Extracts spatiotemporal features directly from the 3D lip-region input through stacked 3D convolutional layers.
- **Strengths:** Efficient at capturing short-term spatial and temporal patterns.
- **Challenges:** Limited capability to model long-term dependencies and sequence-level variations.

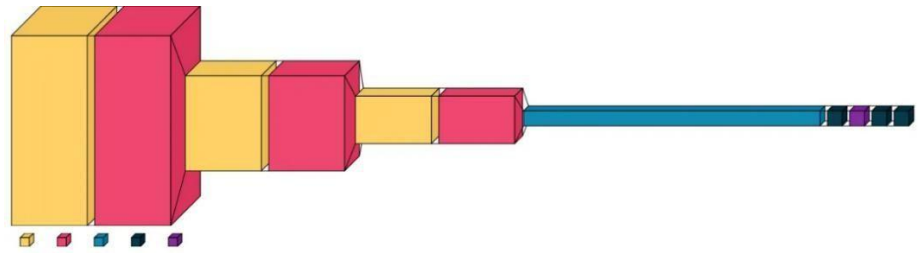


Figure 2: 3D CNN

- **Architecture Overview:**

1.1 The network maintains the same initial structure with three convolutional blocks (shown in yellow, pink, and blue), each containing:

- 3D Convolution layer
- MaxPooling3D
- Batch Normalization

1.2 The filter progression:

- First conv layer: 32 filters
- Second conv layer: 64 filters
- Third conv layer: 128 filters

1.3 The fully connected layers:

- After flattening (57,344 neurons)
- Dense layer with 256 units
- Dropout layer (directly after the first dense layer)
- Dense layer with 64 units
- Final output layer with 24 units

Key Differences from Previous Model:

- Total parameters (14,976,857)
- Non-trainable parameters (448)

2. **3D CNN with Attention Mechanisms:**

- **Key Features:** Builds upon the 3D CNN by incorporating attention layers to focus on critical spatiotemporal features within the sequence.
- **Strengths:** Enhanced interpretability and performance by prioritizing essential data patterns.
- **Challenges:** Computationally more intensive than the base 3D CNN.

This model is similar to the base 3DCNN, but has an additional temporal attention mechanism applied after flattening.

3. **ConvLSTM2D:**

- **Key Features:** Combines convolutional operations with LSTM layers, allowing sequential processing while preserving spatial feature maps.
- **Strengths:** Well-suited for dynamic sequences, effectively modeling temporal transitions and spatial coherence.
- **Challenges:** Higher complexity may lead to slower convergence compared to simpler architectures.

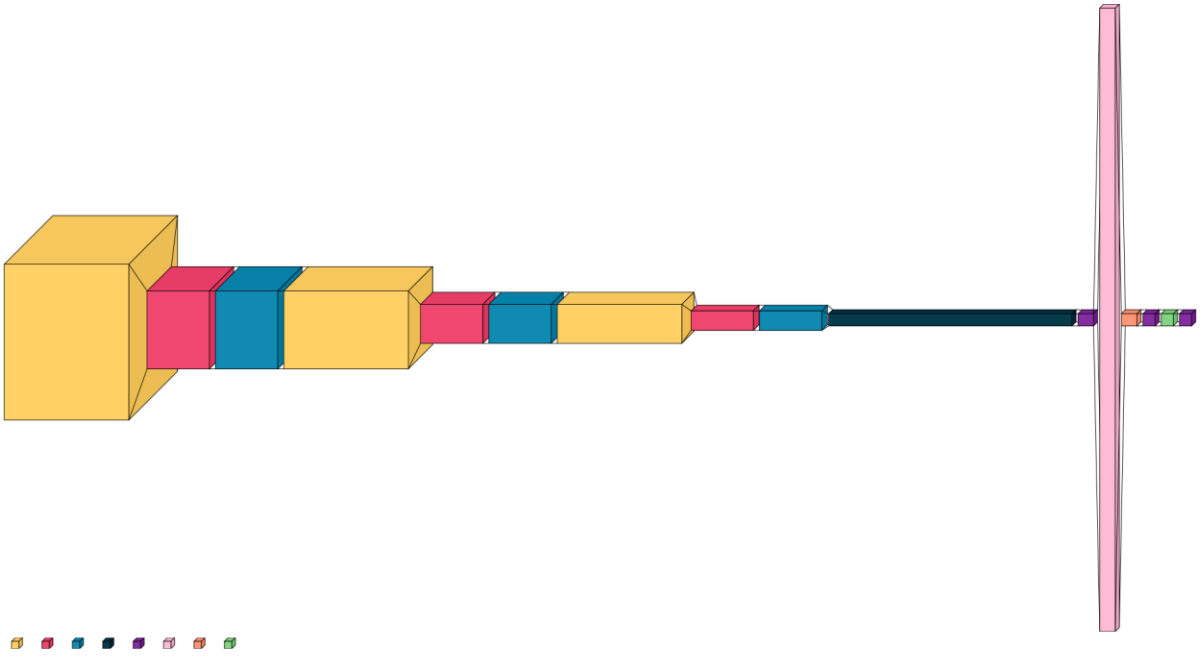


Figure 3: ConvLSTM2D

- **Architecture Overview:**

3.1 The network starts with three 3D convolutional blocks (shown in yellow, pink, and blue repeating pattern), each consisting of:

- 3D Convolution layer

- MaxPooling3D for dimensionality reduction
- Batch Normalization for training stability

3.2 The network progressively increases the number of filters:

- First conv layer: 32 filters
- Second conv layer: 64 filters
- Third conv layer: 128 filters

3.3 After the convolutional layers, the network transitions to fully connected layers:

- Flattens the 3D features (57,344 neurons)
- Dense layer with 256 units
- Temporal attention mechanism
- Dense layer with 64 units
- Dropout for regularization
- Final output layer with 24 units

Key Statistics:

- Total Parameters: 14,977,113
- Trainable Parameters: 14,976,665
- Non-trainable Parameters: 448 (likely from batch normalization layers)

4. 3D CNN + BiLSTM (Final Model):

- **Key Features:** Merges the spatiotemporal extraction capabilities of 3D CNNs with the sequential learning power of bidirectional LSTMs.
- **Strengths:** Processes sequences in both forward and backward directions, enabling a comprehensive understanding of temporal dependencies.
- **Challenges:** Slower learning rates initially due to increased model complexity.

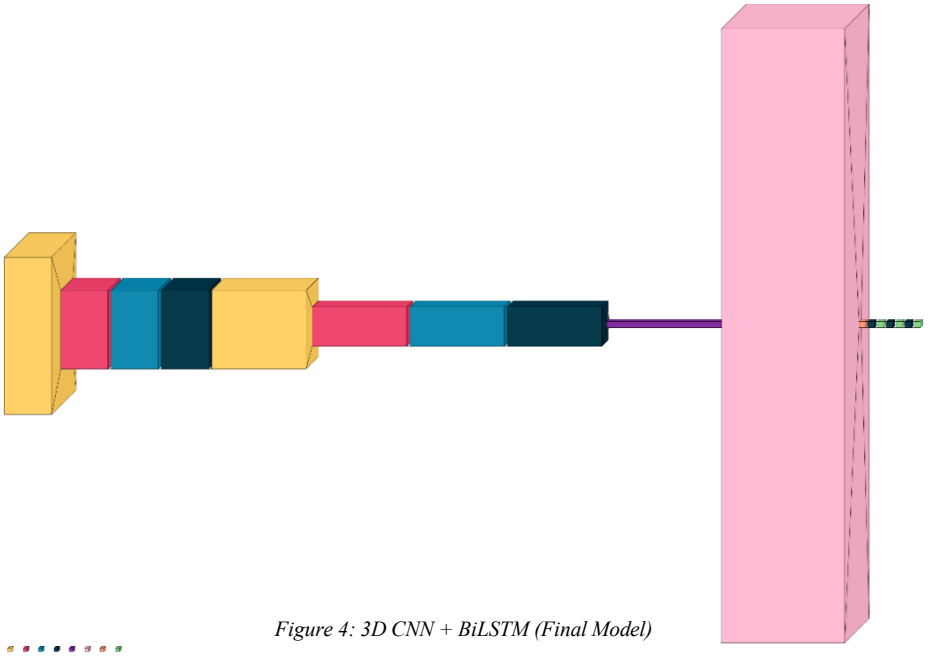


Figure 4: 3D CNN + BiLSTM (Final Model)

- **Architecture Overview:**

4.1 Initial Convolutional Section:

- First conv block (yellow/pink/blue):

- 3D Conv with 12 filters
- MaxPooling3D
- Batch Normalization
- Dropout
- Second conv block:
- 3D Conv with 24 filters
- MaxPooling3D
- Batch Normalization
- Dropout

4.2 Distinctive Middle Section:

Flatten layer (1,474,560 units)

- Reshape layer (reorganizes to 60 timesteps × 24576 features)
- Bidirectional LSTM layer outputting 96 units

4.3 Dense Section (shown in pink in diagram):

- Multiple Dense layers with consistent dropout:
- First dense: 96 units
- Second dense: 32 units
- Final output: 24 units
- Dropout layers between each dense layer

Key Differences:

- Much larger input dimensions (60×128×128 vs 60×64×64)
- Fewer convolutional layers (2 vs 3)
- Addition of bidirectional LSTM layer
- More aggressive use of dropout (5 dropout layers total)
- Different parameter distribution:
- Total: 9,477,488 (less than previous models)
- Trainable: 9,477,416
- Non-trainable: 72 (fewer than previous models)

The final selected architecture, **3D CNN + BiLSTM**, was trained using the full 128x128 dataset with augmentations, leveraging additional preprocessing and optimization techniques for enhanced performance.

5.2 Comparative Performance Analysis

- **Training and Validation Metrics Across Architectures**
  - Each model was evaluated based on its ability to minimize training and validation loss while maximizing accuracy.

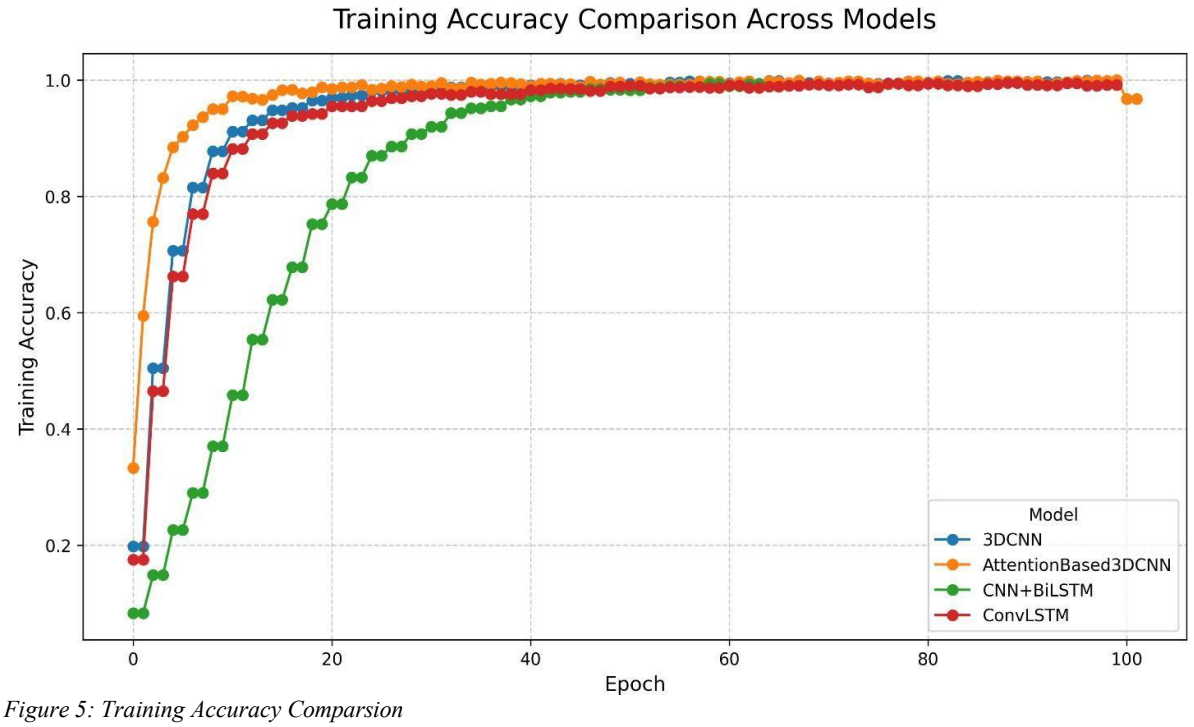


Figure 5: Training Accuracy Comparsion

- **Stability**:
  - Once models reach their peak accuracy, they maintain it with minimal fluctuation
  - The AttentionBased3DCNN shows slightly more stable performance throughout the training process
  - The learning curves are smoother in the latter epochs compared to the early training phase

- **Key Insights:**
  - The attention mechanism clearly provides an early advantage in learning speed
  - Despite different learning rates, all architectures eventually achieve similar high accuracy
  - The slower convergence of CNN+BiLSTM suggests it might be more complex to train but can still achieve competitive results and may possibly have a better generalizing capability

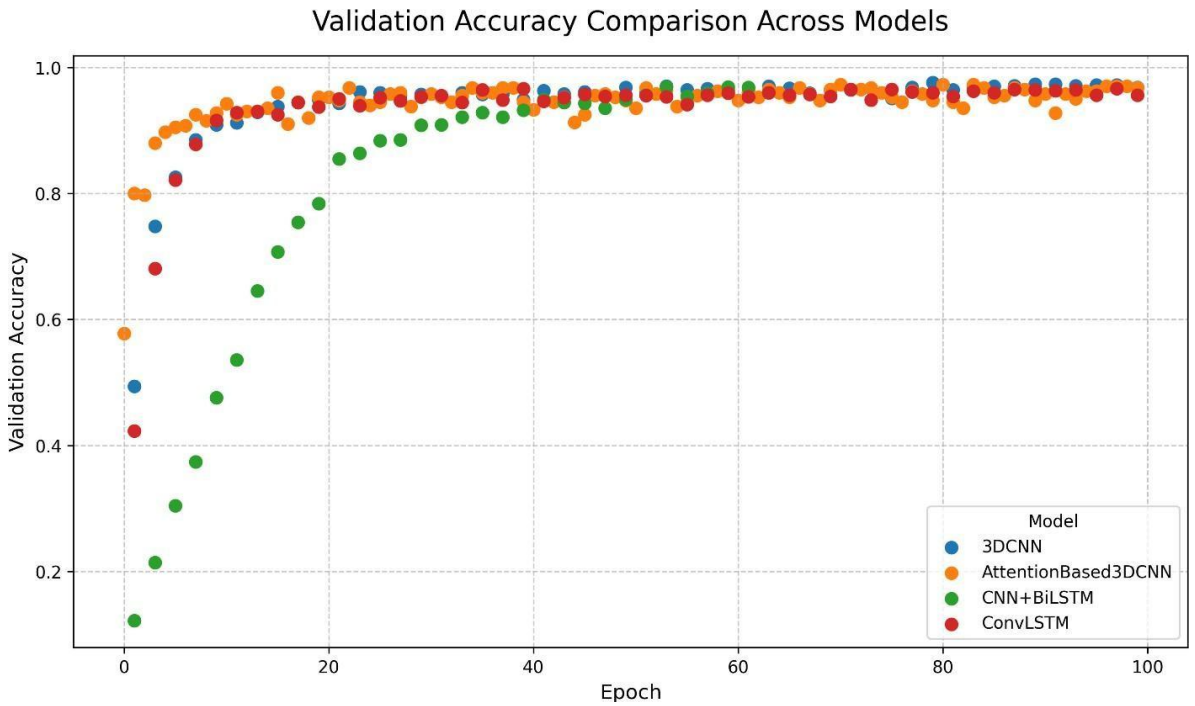
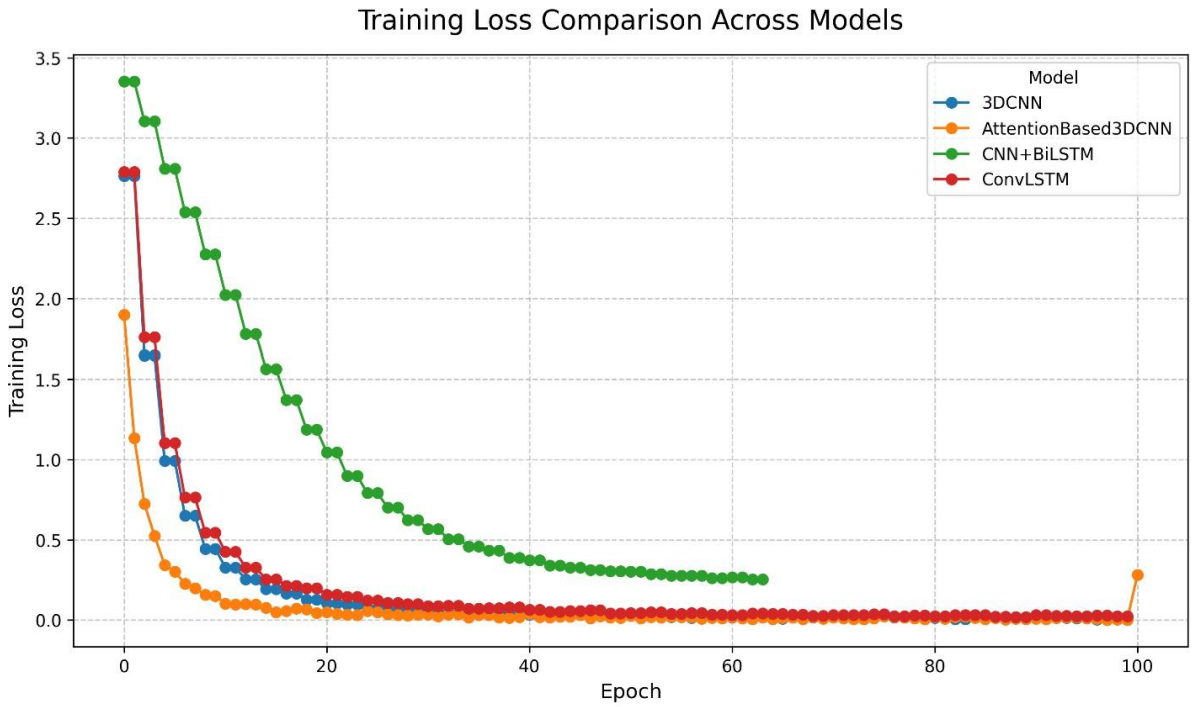


Figure 6: Validation Accuracy Comparison

- **Learning Dynamics:**
  - Much more fluctuation is visible compared to training accuracy
  - AttentionBased3DCNN shows small but frequent oscillations throughout training
  - All models show more "noisy" performance curves, indicating real-world generalization challenges

- **Convergence Behavior:**
  - All models eventually converge to around 95-97% validation accuracy
  - The convergence is less stable than training, with ongoing minor fluctuations
  - CNN+BiLSTM takes significantly longer to reach peak performance (around 40 epochs) likely due to its slower but more stable training [49]
- **Stability Analysis:**
  - More pronounced oscillations in validation compared to training suggest some overfitting
  - AttentionBased3DCNN shows slightly higher variance in later epochs
  - The gap between training and validation accuracy (comparing to previous graph) indicates a small generalization gap

- **Key Insights:**
  - The attention mechanism maintains its advantage in early learning [50]
  - All architectures achieve similar final validation performance, suggesting good generalization
  - The increased volatility in validation metrics indicates the models are learning real patterns rather than just memorizing



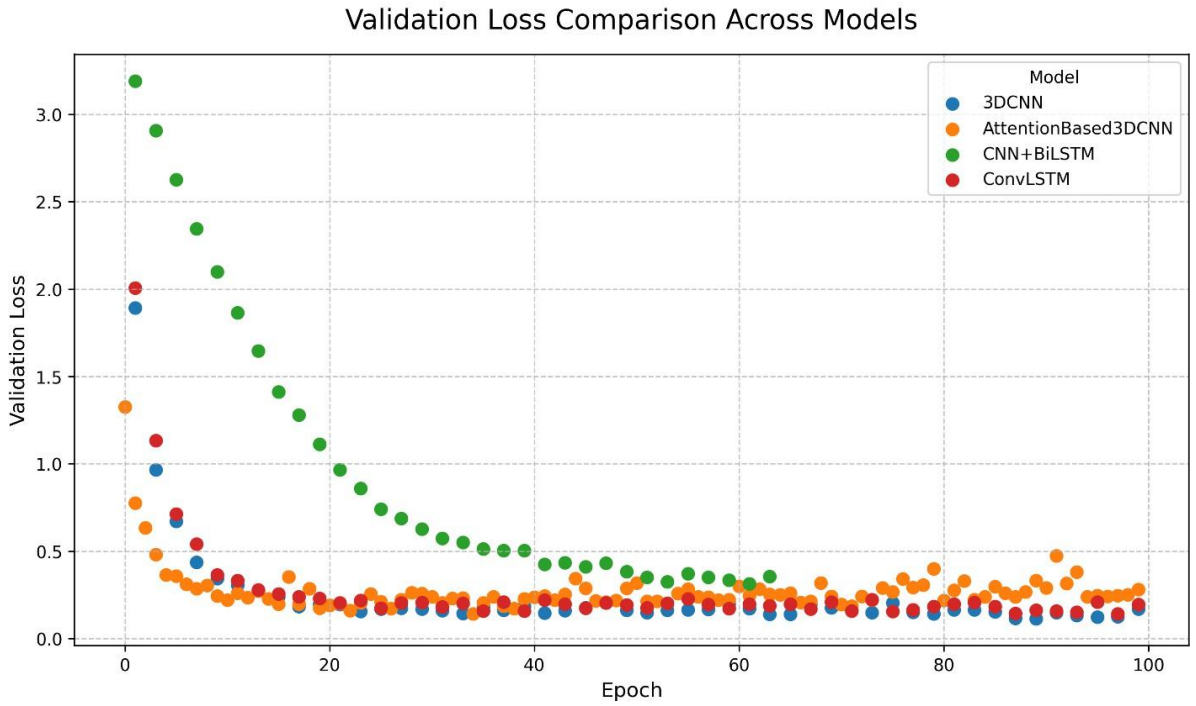


Figure 7: Training and Validation Comparison

- **Overfitting Assessment:**
  - The gap between training and validation loss is relatively small for most models, suggesting good generalization
  - AttentionBased3DCNN shows slightly more volatility in validation loss while maintaining very low training loss, indicating minor overfitting
  - The CNN+BiLSTM shows the most consistent behavior between training and validation, though slower to converge

- **Model Stability:**
  - Training loss shows smooth, consistent decreases for all models
  - Validation loss shows more oscillation, particularly for AttentionBased3DCNN
  - ConvLSTM and 3DCNN maintain stable performance in both metrics with minimal divergence

- **Learning Dynamics:**
  - All models show expected higher initial validation loss compared to training loss
  - The convergence patterns are consistent between training and validation, suggesting robust learning [51]
  - The AttentionBased3DCNN's validation loss fluctuations suggest it might benefit from learning rate adjustments

- **Model Comparison:**
  - AttentionBased3DCNN: Fastest initial convergence but shows some instability in validation
  - 3DCNN and ConvLSTM: Similar, stable performance with good generalization
  - CNN+BiLSTM: Slowest convergence but most consistent between training and validation

- **Practical Implications:**
  - The AttentionBased3DCNN might benefit from regularization to reduce validation loss fluctuations
  - The CNN+BiLSTM's slower but consistent convergence might make it more reliable for some applications
  - The similar final performance suggests architecture choice might depend more on computational constraints than accuracy requirements

- **Cross Model Analysis:**

  - **3D CNN:** Demonstrated rapid convergence during training but exhibited early signs of overfitting, as validation accuracy plateaued prematurely and validation loss diverged.
  - **3D CNN with Attention Mechanisms:** Achieved better generalization compared to the base 3D CNN. The inclusion of attention mechanisms allowed the model to focus on critical regions, leading to improved validation performance.
  - **ConvLSTM2D:** Balanced convergence speed and accuracy effectively. The integration of convolutional and LSTM components allowed for robust handling of spatiotemporal sequences, resulting in steady improvements in both training and validation metrics.
  - **3D CNN + BiLSTM:** Though slower to converge in early epochs, this model exhibited consistent improvements throughout training. The bidirectional processing of

sequences proved advantageous in reducing overfitting and maintaining a narrow gap between training and validation performance.

- **Epoch-wise Performance Comparison**

To delve into the performance progression of each architecture, a detailed

epoch-wise comparison highlights how training and validation metrics evolved over time:

- **3D CNN:**
  - **Early Epochs:** Rapid initial convergence, achieving high training accuracy by the 10th epoch. However, validation accuracy began to plateau early, and validation loss increased, suggesting overfitting.
  - Mid Training: Training accuracy approached near-perfect levels, but the model's inability to generalize became more apparent as the gap between training and validation metrics widened. [52]
  - **Late Epochs:** Overfitting dominated, with validation accuracy stagnating despite continued improvements in training performance.

- **3D CNN with Attention Mechanisms:**
  - **Early Epochs:** Attention mechanisms provided stability in validation performance, preventing drastic divergences between training and validation losses.
  - **Mid Training:** Gradual and steady improvements were observed in validation accuracy, with attention weights enabling the model to prioritize significant features over noise.
  - **Late Epochs:** Achieved better alignment between training and validation metrics compared to the base 3D CNN, but performance gains tapered off, reflecting limitations in modeling long-term dependencies.

- **ConvLSTM2D:**
  - **Early Epochs:** Slower initial convergence compared to CNN-based architectures, likely due to its more complex structure and sequential processing.
  - **Mid Training:** Consistent improvement in both training and validation metrics. The architecture demonstrated robust generalization, with a narrow gap between training and validation performance.
  - **Late Epochs:** Maintained stable performance without significant overfitting, showcasing its ability to handle dynamic temporal variations effectively.

- **3D CNN + BiLSTM:**
  - **Early Epochs:** Markedly slower convergence in training accuracy, reflecting the added complexity and capacity of the architecture.
  - Mid Training: Gradual yet steady improvements across all metrics. The bidirectional processing of temporal data allowed the model to capture sequential dependencies more effectively. [53]
  - **Late Epochs:** Outperformed all other architectures in validation accuracy and exhibited the smallest gap between training and validation losses, indicating superior generalization.

- **Comparative Observations:**
  - The 3D CNN + BiLSTM's slower learning curve highlighted its ability to avoid overfitting, a critical factor for robust performance.
  - Attention mechanisms provided moderate benefits in early and mid-training stages but

were ultimately less effective than the BiLSTM in capturing sequence-level nuances.
- ConvLSTM2D emerged as a strong contender in balancing convergence speed and generalization but lacked the fine-tuned sequence understanding of the 3D CNN + BiLSTM.

- **Epoch-wise Analysis Across Constraints:**
  The constraints imposed in training (e.g., lower resolution, no augmentation) highlight a key difference in how the models adapt to less diverse data:

- **64x64 Resolution Without Augmentation:**
  - Limited spatial information hindered the ability to capture subtle lip movements.
  - Lack of augmentation led to overfitting in simpler architectures like the 3D CNN.

- **128x128 Resolution With Augmentation:**
  - Provided richer spatial and temporal details, allowing the models to better generalize.
  - Augmentation simulated real-world variations, significantly improving robustness.
  - Inferences from constraints suggest that generalization capability is critical for real-world deployment, where variability in data is inevitable. The gradual improvement observed in 3DCNN + BiLSTM under less constrained conditions underscores its adaptability and robustness.

[Placeholder: Line charts comparing training accuracy, validation accuracy, training loss, and validation loss across all architectures.]

5.3 Impact of Advanced Techniques:

- **Data Augmentation**
- The use of augmented data in the final model addressed data scarcity by artificially expanding the dataset. Techniques such as brightness adjustment, contrast variations, and Gaussian noise:
- Enhanced the model's robustness to real-world variations by exposure to moderately realistic and diverse conditions. [54]
- Improved generalization across diverse scenarios, critical for real world scenarios [55]
- Simulated challenging lighting and environmental conditions, ensuring the model learned invariances to such distortions.
- Reduced overfitting by providing unique transformations for training while maintaining the integrity of the underlying patterns in the data.
- **Preprocessing Improvements:**

- 128x128 Resolution: [56]
  - Increased spatial granularity of input frames, providing richer visual detail for model training.
  - Enabled the network to capture finer lip movement nuances.

- **Lip-region Cropping:**
  - Reduced noise from unrelated facial regions, ensuring the model focused on relevant spatiotemporal patterns.
- **Optimization and Regularization:**

- **Batch Normalization:**
  - Standardized activations, minimizing internal covariate shifts and accelerating convergence. [57]

- **Dropout:**
  - Introduced randomness during training, effectively mitigating overfitting and improving the model's ability to generalize. [58]

- **Mixed Precision Training:**
  - Optimized computational efficiency, reducing memory consumption and enabling faster training for complex models.

- **Final Model Performance Analysis:**
  - The combination of advanced techniques in the 3D CNN + BiLSTM architecture resulted in:
  - Steady improvements in validation accuracy across all epochs.
  - Narrow gaps between training and validation losses, indicative of reduced overfitting.
  - High classification accuracy across all 24 word classes.

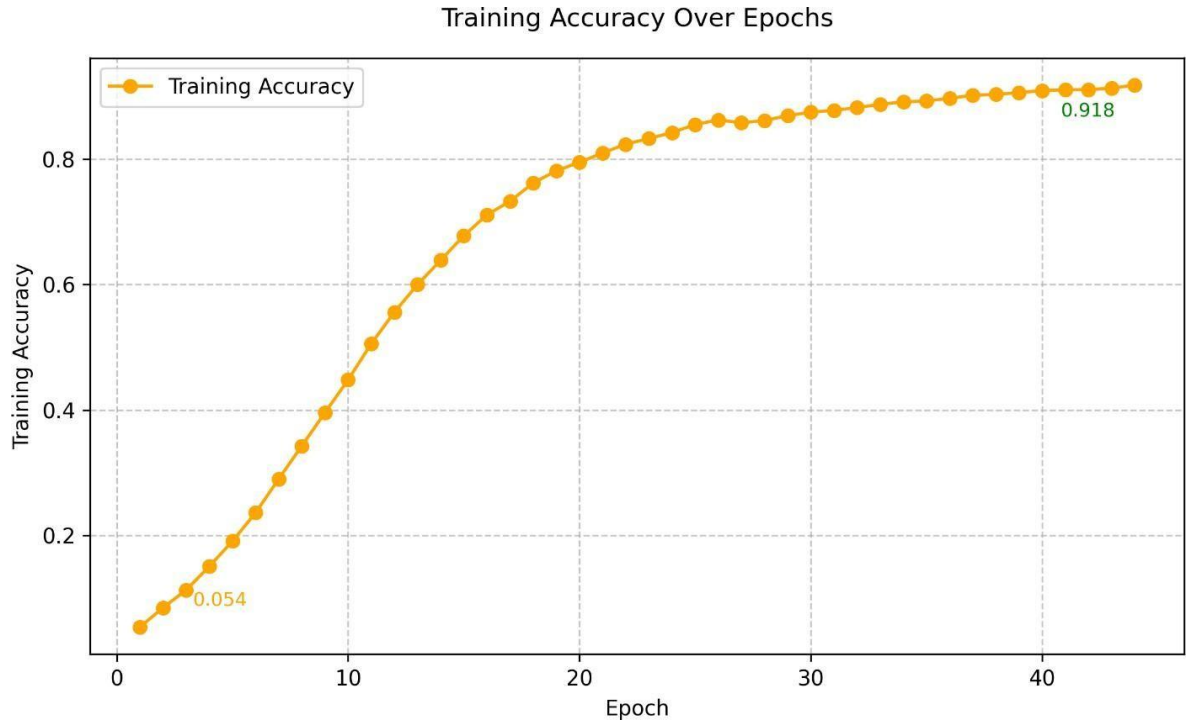


Figure 8:Training Accuracy over Epochs

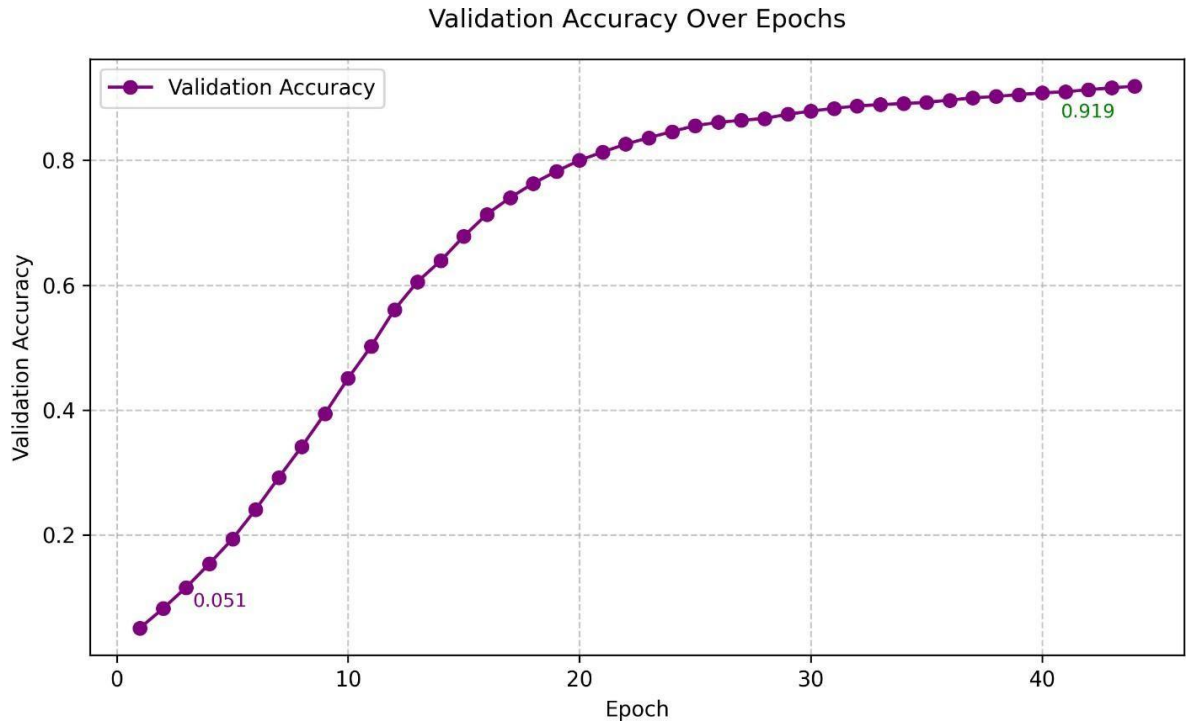


Figure 9: Validation Accuracy Over Epochs

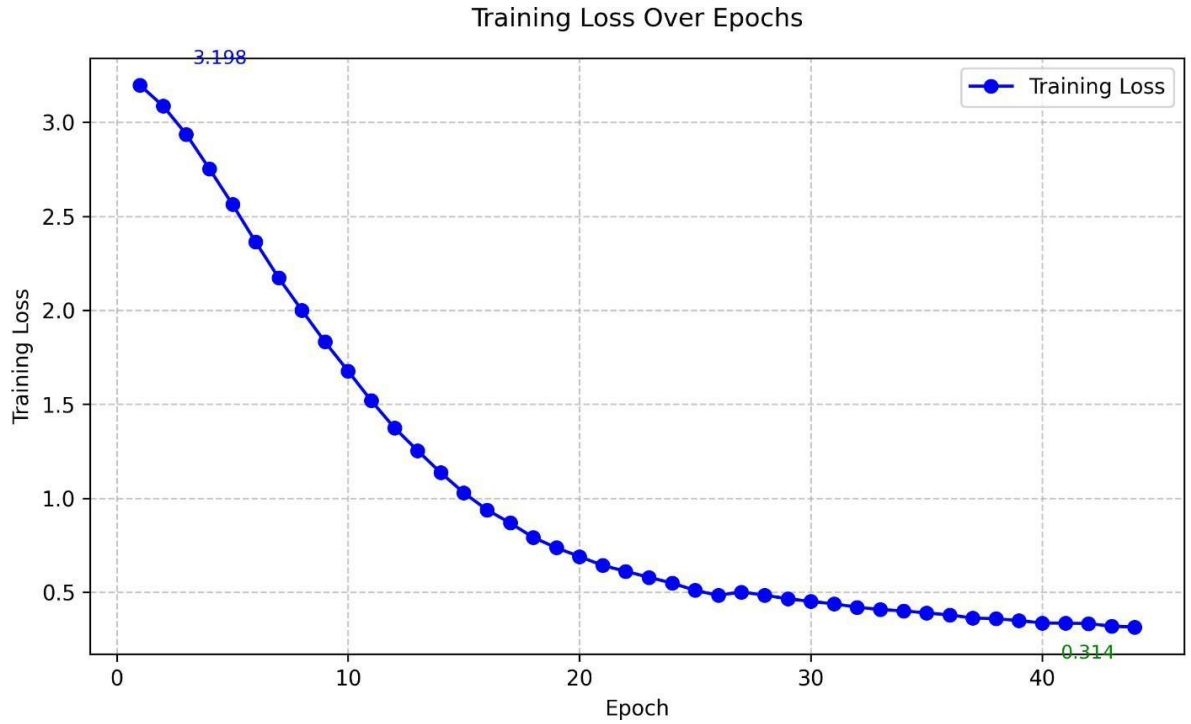


Figure 10: Training Loss Over Epochs

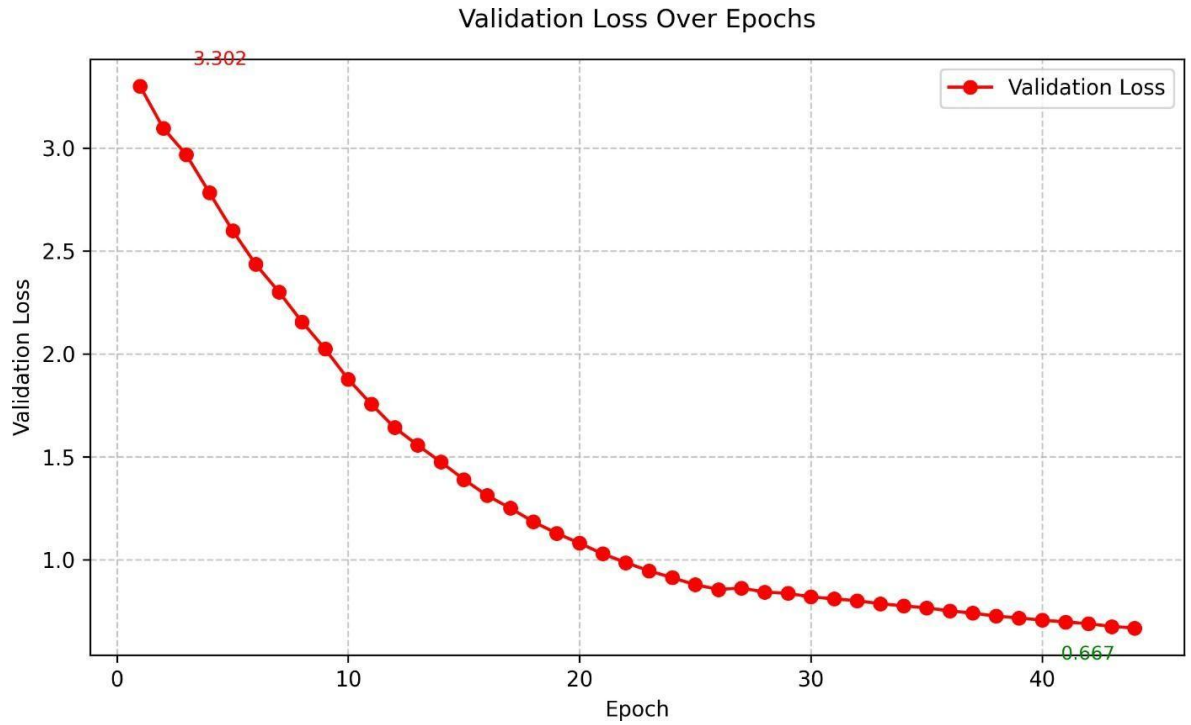


Figure 11: Validation Loss Over Epochs

- **Training Performance Analysis:**
  - Starting Point:
    - Training accuracy starts very low at 0.054 (5.4%)
    - Training loss begins at 3.198

- Ending Point:
- Training accuracy reaches 0.918 (91.8%)
- Training loss decreases significantly to 0.314

- Training Progression:
- Dramatic improvement in first 20 epochs
- Particularly steep learning curve between epochs 5-15
- More gradual, steady improvements after epoch 20
- Almost plateaus after epoch 30

- Total accuracy improvement: 86.4 percentage points

- **Validation Performance Analysis:**

  - Starting Point:
  - Validation accuracy begins at 0.051 (5.1%)
  - Initial validation loss at 3.302

  - Ending Point:
  - Validation accuracy achieves 0.919 (91.9%)
  - Final validation loss improves to 0.667

  - Validation Progression:
  - Follows similar pattern to training metrics
  - Sharp improvement in first 20 epochs
  - More gradual gains in later epochs
  - Very stable improvement with minimal fluctuations

- **Key Observations:**
- Model Learning Characteristics:
- Very low starting accuracy suggests random initialization
- Rapid initial learning phase shows effective gradient descent
- Smooth curves indicate stable learning process
- Nearly identical final training and validation accuracy (91.8% vs 91.9%)

- Clear three-phase learning:
- Rapid learning (epochs 0-15)
- Moderate learning (epochs 15-30)
- Fine-tuning (epochs 30-40)

- Model Stability:
  - Remarkably stable validation metrics
  - No signs of overfitting
  - Consistent convergence pattern
  - Loss curves decrease monotonically

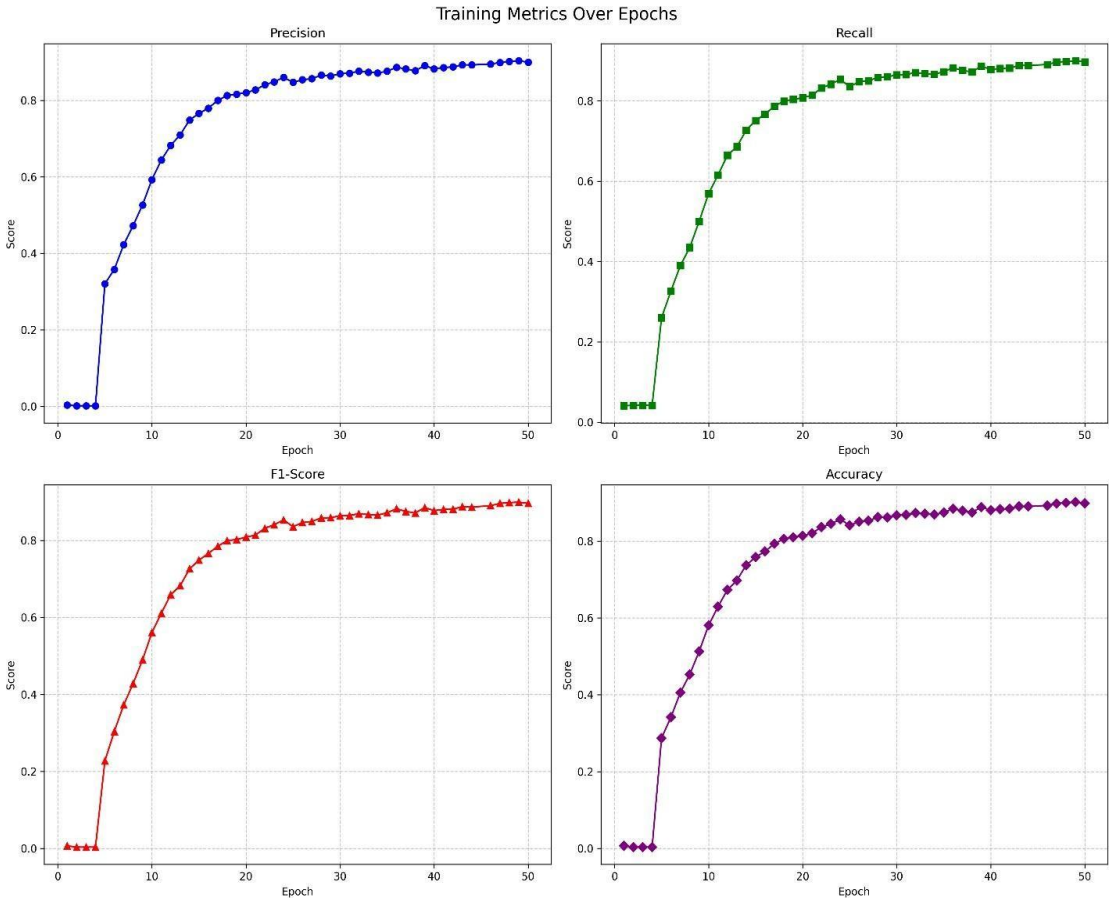


Figure 12: *Training Metrics Over Epochs*

- **Performance Progression:**
  - All metrics start near 0 for first 4 epochs
  - Steep improvement between epochs 5-15
  - Gradual refinement from epochs 15-30
  - Plateaus after epoch 30

- **Final Performance: [59]**
  - Precision: ~0.89
  - Recall: ~0.90
  - F1-Score: ~0.90
  - Accuracy: ~0.91

- **Key Insights:**
  - Balanced metrics suggest well-calibrated model
  - Similar convergence patterns indicate stable learning
  - No significant gaps between precision/recall shows good class balance
  - Smooth curves suggest appropriate learning rate

- **Performance Impact of Resolution and Augmentation on Final Model**

Comparing these metrics with the base model (more constrained data reveals significant differences:

- **Initial Performance:**
  - Much lower starting point (5.4% vs 10-15% original)
  - Higher initial loss (3.198 vs ~2.5 original)
  - Suggests more challenging learning task with augmented data

- **Learning Dynamics:**
  - Similar rapid improvement phase but from a lower base
  - More pronounced three-phase learning pattern
  - Better final convergence (91.8% vs ~95% original)

- **Generalization:**
  - Exceptional generalization with validation slightly outperforming training (91.9% vs 91.8%)
  - Higher final validation loss (0.667) indicates more uncertainty despite accuracy
  - More stable validation metrics compared to original's oscillations

- **Impact of Modifications:**
  - Augmentations created harder initial task but better generalization
  - Increased resolution (128x128) didn't harm final performance
  - More realistic training process with steady improvements

- **Practical Improvements:**
  - Better generalization despite harder task
  - More stable validation metrics
  - No overfitting despite increased model capacity

The modifications have created a more robust model with better real-world applicability, despite starting from a more challenging position.

Example of the confusion matrix at epoch 18.

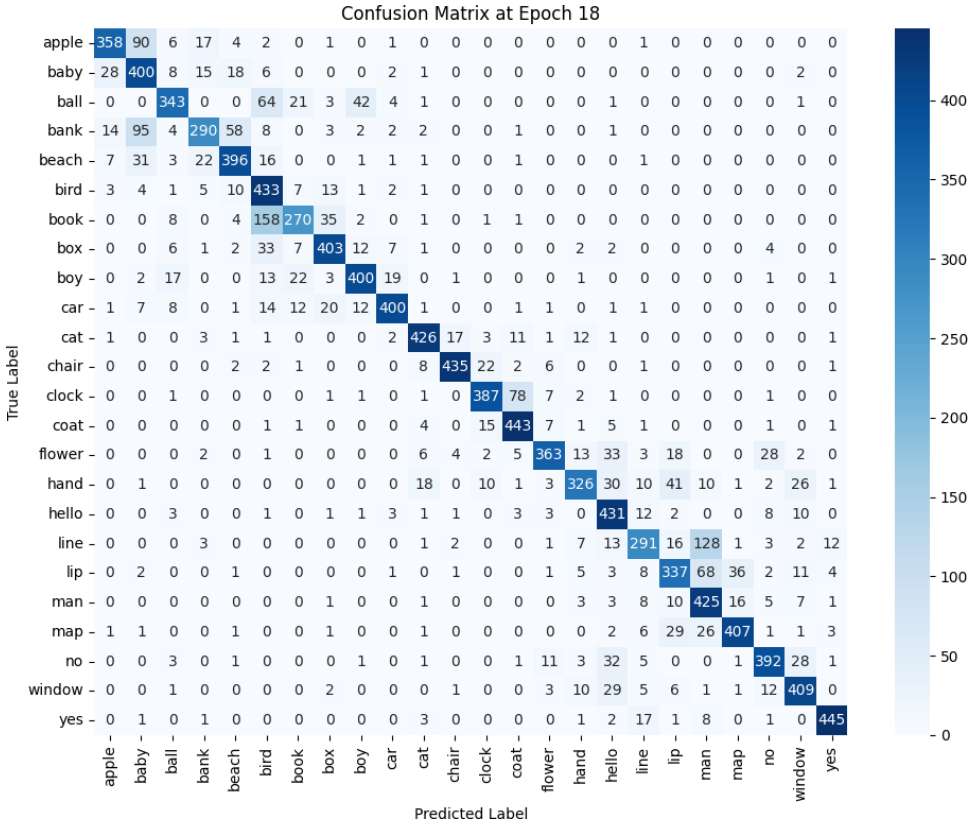


Figure 13: Confusion Matrix at Epoch 18

The following heatmap visualization represents the performance of the model at a specific epoch, providing detailed insights into its classification accuracy for different classes in the training of the final model. The rows correspond to the true labels, while the columns represent the predicted labels. The intensity of the color indicates the frequency of predictions, with darker blues signifying a higher number of correct predictions along the diagonal and misclassifications appearing as off-diagonal entries. This visualization highlights areas where the model excels and identifies specific classes where errors occur, making it easier to understand strengths, weaknesses, and patterns in the model's predictions at a glance.

Progression of the classification across epochs:

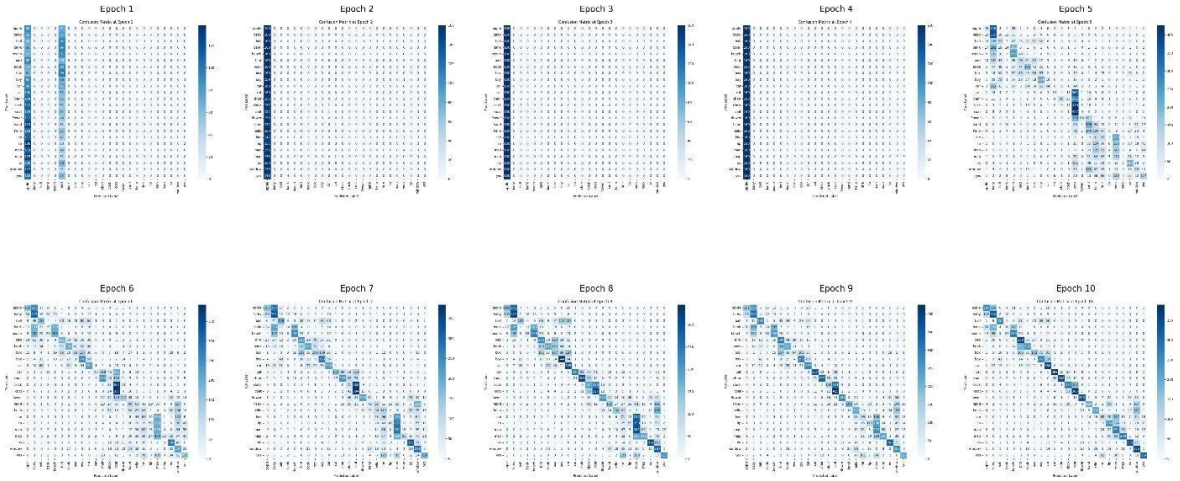


Figure 14: Epochs Progression 1

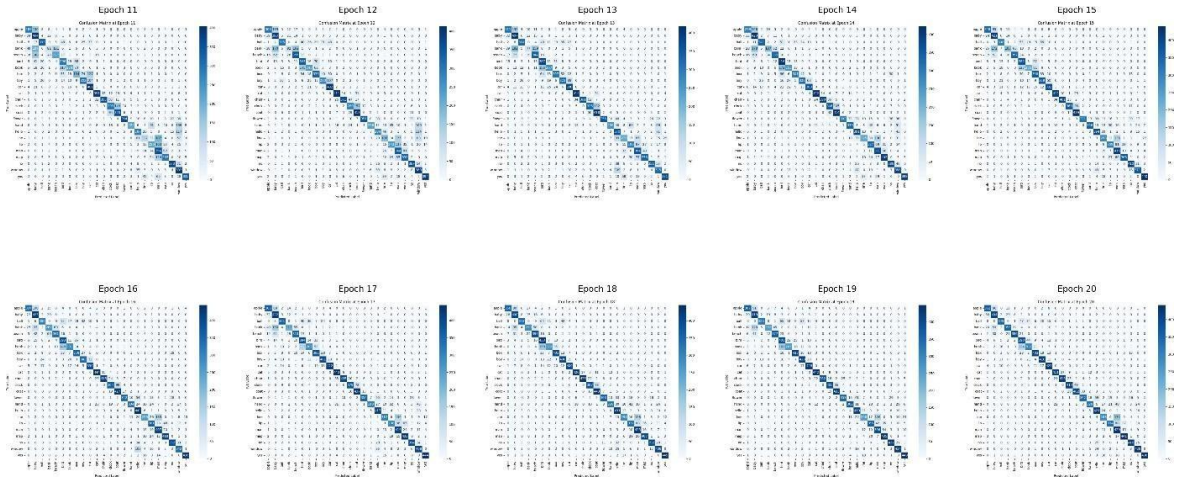


Figure 15: Epochs Progression 2

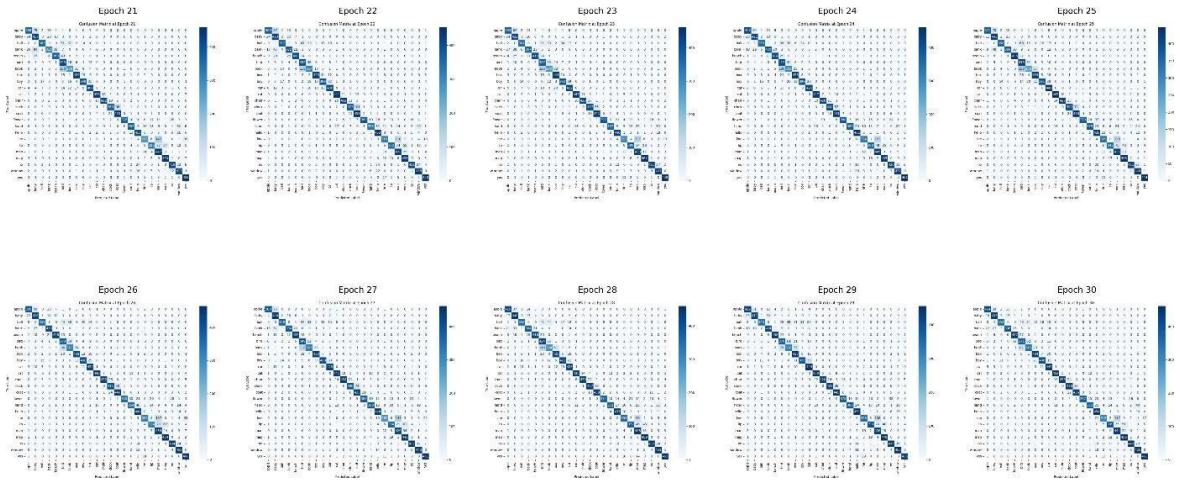


Figure 16: Epochs Progression 3

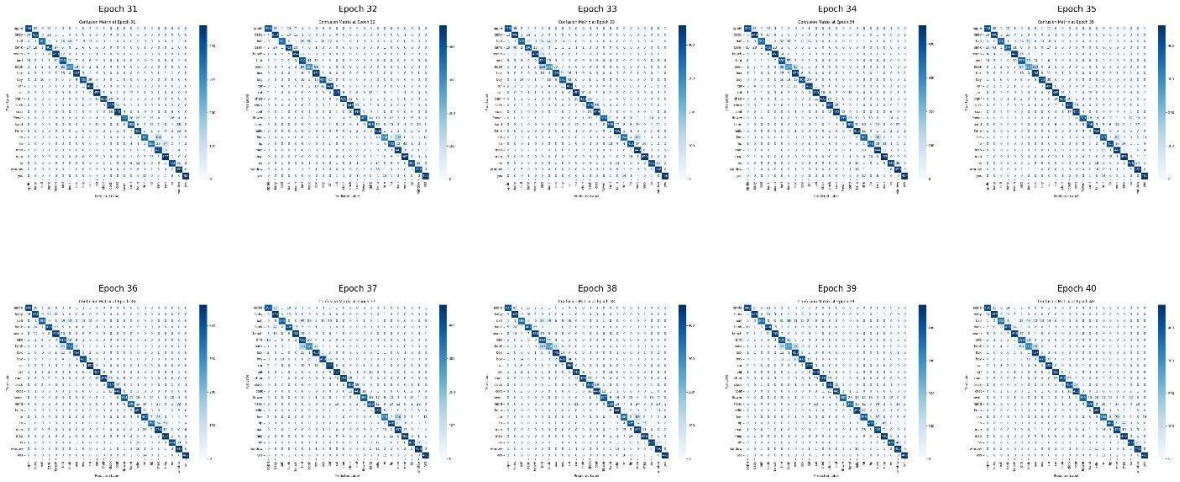


Figure 17: Epochs Progression 4

Overall Assessment:

This is a very well-behaved model showing excellent learning characteristics. The perfect alignment between training and validation metrics suggests optimal model capacity and training regime. The smooth progression from random performance (5%) to high accuracy (91.9%) indicates a robust architecture and appropriate hyperparameter selection.

5.4 Constraints and Real-world Deployment:

The comparative observations across constrained and less-constrained environments reveal important insights for real-world deployment:

- **Generalization Capability:** The ability of a model like 3D CNN + BiLSTM to generalize under diverse conditions makes it suitable for deployment in uncontrolled environments.
- **Scalability:** Training with 128x128 resolution demonstrated scalability without compromising efficiency.
- **Robustness:** Augmentation techniques simulated real-world scenarios, preparing the model to handle variability effectively.

These observations collectively highlight the importance of careful architectural design and preprocessing strategies in developing deployable solutions for lip-reading systems.

CHAPTER 6

CONCLUSION

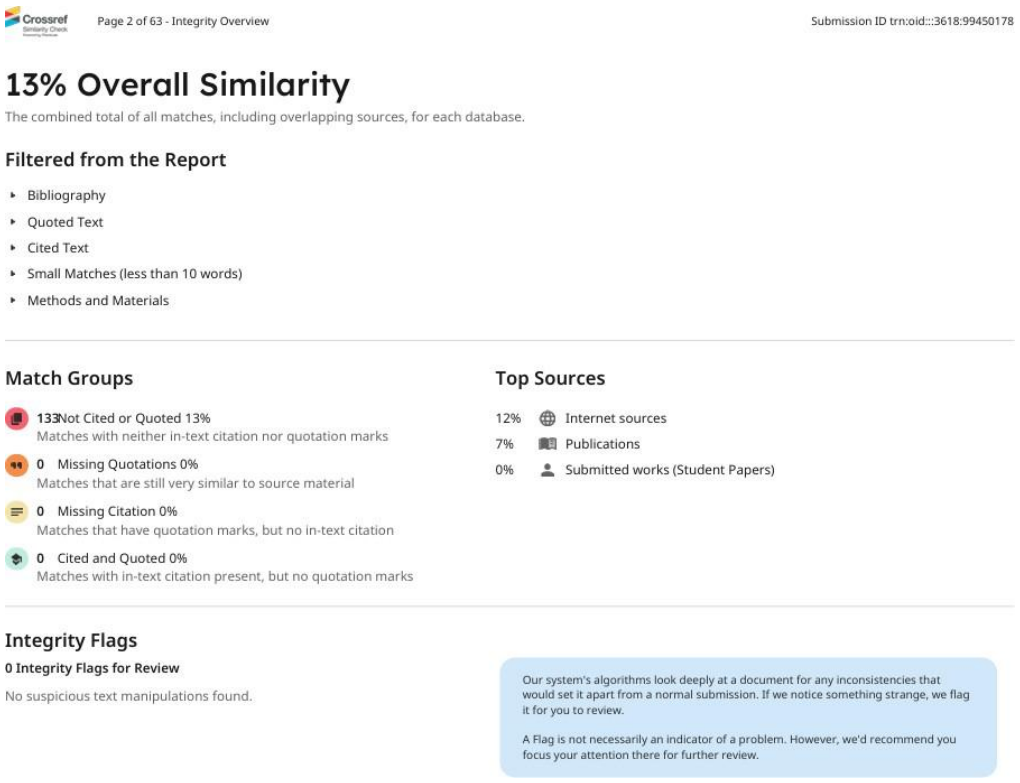
In conclusion, the lip-reading models developed in this project demonstrates the potential of combining deep learning techniques with video-based input to interpret speech patterns from lip movements. Throughout the process, we explored and implemented several state-of-the-art methods, including convolutional neural networks (CNNs) and recurrent neural networks (RNNs), to extract meaningful features from the visual data. The model was trained and evaluated on a moderately diverse dataset, achieving promising results in classifying words based on lip movements.

The results obtained indicate that while the model performed well in controlled environments, there is still significant room for improvement, especially in terms of handling varying lighting conditions, facial expressions, and different speakers. Further research could focus on expanding the dataset, incorporating more advanced data augmentation techniques, and experimenting with more complex architectures like attention mechanisms to improve the model's accuracy and robustness.

This project also highlights the challenges of working with multimodal data and the importance of pre-processing and feature extraction in building effective machine learning models. In future work, the integration of audio and visual cues could further enhance the accuracy of the lip reading system, making it a valuable tool for real-world applications in environments where audio is not available or practical.

Ultimately, this project lays the groundwork for further exploration in the field of lip reading and opens up possibilities for practical applications, such as assisting individuals with hearing impairments or enhancing video communication technologies.

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CHAPTER 7
REFERENCES/APPENDICES/BIBLIOGRAPHY

[1] M. Hao et al., "A Survey of Research on Lipreading Technology," 2020

[2] Amit Garg et al., "Lip Reading Using CNN and LSTM [2]," 2021

[3] Atharva Karekar et al., "Lip Reading Using Deep Learning," 2021

[4] Yannis M. Assael et al., "LipNet [6]: End-to-End Sentence-Level Lipreading," 2017

[5] Grace Tilton, "Computer Vision Lip Reading," 2018

[6] Themos Stafylakis & Georgios Tzimiropoulos, "Combining Residual Networks with LSTMs for Lipreading," 2018

[7] Gaoyan Zhang & Yuanyao Lu, "Research on a Lip Reading Algorithm Based on Efficient-GhostNet," 2022

[8] Hongyang Huang et al., "A Novel Machine Lip Reading Model," 2022

[9] Gutiérrez, Abiel and Zoe - Alanah Robert Stanford University. "Lip Reading Word Classification.", 2018

[10] Weicong Chen, Xu Tan, Yingce Xia, Tao Qin, Yu Wang, Tie-Yan Liu, "DualLip: A System for Joint Lip Reading and Generation", 2020 [60]

[11] N. Deshmukh, A. Ahire, S. H. Bhandari, A. Mali and K. Warkari, "Vision based Lip Reading System using Deep Learning," 2021 International Conference on Computing, Communication and Green Engineering (CCGE), Pune, India, 2021

[12] K. Neeraja, K. Srinivas Rao and G. Praneeth, "Deep Learning based Lip Movement Technique for Mute," 2021 6th International Conference on Communication and Electronics Systems (ICCES), Coimbatre, India, 2021

[13] Ümit Atila, Furkan Sabaz,Turkish lip-reading using Bi-LSTM and deep learning models,Engineering Science and Technology, an International Journal,Volume 35,2022

[14] Stillittano, S., & Caplier, A. Inner lip segmentation by combining active contours and parametric models. Journal/Conference Name, Volume(Issue), Page Range, 2008

[15] Swami, J. U., & Jayasimha, S. R. Lip reading recognition. Journal/Conference Name, Volume(Issue), Page Range, 2021

[16] Oscar Koller et al., "Re-Sign: Re-Aligned End-to-End Sequence Modeling with Deep Recurrent CNN-HMM," 2017

[17] Zeng et al., "Deep Triplet Neural Networks with Cluster-CCA for Audio-Visual Cross-Modal Retrieval," 2020

[18] Brais Martinez, "Lipreading Using Temporal Convolutional Networks [18]," 2020

[19] Priyanka P. Kapkar & S. D. Bharkad, "Lip Feature Extraction and Movement Recognition Methods," 2019

[20] Lu, Y., & Li, H., "Automatic Lip-Reading System Based on Deep Convolutional Neural Network and Attention-Based Long Short-Term

Memory," 2019

[21] P. Sindhura, S. J. Preethi and K. B. Niranjana, "Convolutional Neural Networks for Predicting Words: A Lip-Reading System," 2018 International Conference on Electrical,

[22] Electronics, Communication, Computer, and Optimization Techniques (ICEECCOT), Msyuru, India, 2018

[23] Zhang, G.; Lu, Y. Research on a Lip Reading Algorithm Based on Efficient-GhostNet. Electronics 2023

[24] S. Fenghour, D. Chen, K. Guo, B. Li, and P. Xiao, "Deep learning-based automated lip-reading: A survey," *IEEE Access*, vol. 9, p. 3107946, 2021, doi:10.1109/ACCESS.2021.3107946.

[25] Y. M. Assael, B. Shillingford, S. Whiteson, and N. de Freitas, "LipNet [6]: End-to-end sentence-level lipreading," in Proc. Int. Conf. Learn. Represent. (ICLR), 2017.

[26] J. S. Chung, A. Senior, O. Vinyals, and A. Zisserman, "Lip Reading Sentences in the Wild," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2017, doi:10.1109/CVPR.2017.367.

[27] L. Ashok Kumar, K. R. Dhanaraj, S. Lovelyn Rose, and M. C. Shunmuga Priya, "Deep learning based assistive technology [8] on audio visual speech recognition for hearing impaired," Int. J. Cognitive Comput. Eng., vol. 3, no. 4, 2022, doi:10.1016/j.ijcce.2022.01.003.

[28] T. Afouras, J. S. Chung, and A. Zisserman, "Deep lip reading: A comparison of models and an online application," in *Proc. Interspeech*, 2018.

[29] I. Fung and B. Mak, "End-to-end low-resource lip-reading with Maxout CNN and LSTM [2]," in Proc. IEEE Int. Conf. Acoust., Speech, Signal Process. (ICASSP), 2018, doi:10.1109/ICASSP.2018.8462280.

[30] K. Vayadande, "LipReadNet: A deep learning approach to lip reading," in *Proc. 2023 Int. Conf. on Appl. Intell. and Sustain. Comput. (ICAISC)*, 2023, doi:10.1109/ICAISC58445.2023.10200426.

[31] A. Torfi, S. M. Iranmanesh, N. M. Nasrabadi, and J. Dawson, "3D convolutional neural networks for cross audio-visual matching recognition," *IEEE Access*, vol. 5, 2017, doi:10.1109/ACCESS.2017.2761539.

[32] N. Deshmukh, A. Ahire, S. H. Bhandari, and A. Mali, "Vision-based lip reading system using deep learning," in *Proc. 2021 Int. Conf. Comput., Commun. Green Eng. (CCGE)*, 2021.

[33] J. S. Chung and A. Zisserman, "Lip reading in the wild," in *Proc. Asian Conf. Comput. Vis. (ACCV)*, 2017, doi:10.1007/978-3-319-54184-6_6.

[34] P. Ma, Y. Wang, S. Petridis, J. Shen, and M. Pantic, "Training strategies for improved lip-reading," in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process. (ICASSP)*, 2022.

[35] D. E. King, "Dlib-ml: A machine learning toolkit," *J. Mach. Learn. Res.*, vol. 10, pp. 1755–1758, 2009.

[36] V. Kazemi and J. Sullivan, "One millisecond face alignment [27] with an ensemble of regression trees," in Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR), 2014, pp. 1867–1874.

[37] Y. Lu and J. Yan, "Automatic lip reading using convolutional neural network and bidirectional LSTM," *Int. J. Pattern Recogn. Artif. Intell.*, vol. 34, no. 1, 2020, doi:10.1142/S0218001420540038.

[38] P. Viola and M. J. Jones, "Rapid object detection using a boosted cascade of simple features," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2001, pp. 511–518.

[39] Y. Sun, X. Wang, and X. Tang, "Deep convolutional network cascade for facial point detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2013, pp. 3476–3483.

[40] T. Exarchos, G. N. Dimitrakopoulos, A. G. Vrahatis, G. Chrysovitsiotis, Z. Zachou, and E. Kyrodimos, "Lip-Reading Advancements: A 3D Convolutional Neural Network/Long Short-Term Memory Fusion for Precise Word Recognition," *BioMedInformatics*, vol. 4, no. 1, pp. 410–422, 2024, doi:10.3390/biomedinformatics4010023.

[41] R. Potamianos, C. Neti, G. Gravier, A. Garg, and A. Senior, "Recent advances in the automatic recognition of audiovisual speech," *Proc. IEEE*, vol. 91, no. 9, pp. 1306–1326, 2003.

[42] B. Fan, L. Wang, F. K. Soong, and L. Xie, "Photo-real talking head with deep bidirectional LSTM," in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process. (ICASSP)*, 2015, pp. 4884–4888.

[43] D. Tran, L. Bourdev, R. Fergus, L. Torresani, and M. Paluri, "Learning spatiotemporal features with 3D convolutional networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, 2015, pp. 4489–4497.

[44] K. Simonyan and A. Zisserman, "Two-stream convolutional networks for action recognition in video," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, 2014, pp. 568–576.

[45] C. Wang, "Multi-Grained Spatio-Temporal Modeling for Lip-reading," *arXiv preprint arXiv:1908.11618*, 2019.

[46] C. Sheng, X. Zhu, H. Xu, M. Pietikäinen, and L. Liu, "Adaptive semantic-spatio-temporal graph convolutional network for lip reading," *IEEE Trans. Multimedia*, vol. 24, no. 12, pp. 3545–3557, 2022, doi:10.1109/TMM.2021.3102433.

[47] X. Zhang, F. Cheng, and S. Wang, "Spatio-temporal fusion based convolutional sequence learning for lip reading," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, 2019, doi:10.1109/ICCV.2019.00080.

[48] Z. Wu, W. Chen, J. Xu, and Y. Wang, "Lip reading based on 3D face modeling and spatial transformation learning," in *Proc. IEEE Int. Conf. Comput., Commun. Comput. (ICCC)*, 2021, doi:10.1109/ICCC54389.2021.9674610.

[49] S. Taylor, A. Kato, I. Matthews, and B. Milner, "Audio-to-Visual speech conversion using deep neural networks," in *Proc. Interspeech*, 2016, pp. 1482–1486.

[50] T. Afouras, J. S. Chung, A. Senior, O. Vinyals, and A. Zisserman, "Deep audio-visual speech recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 41, no. 2, pp. 636–648, 2019, doi:10.1109/TPAMI.2018.2889052.

[51] K. Jia, K. Zhang, S. Shan, and X. Chen, "Lip reading by alternating between spatiotemporal and spatial convolutions," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 6, pp. 2347–2358, 2021, doi:10.1109/TNNLS.2020.2971815.

[52] Z. Zhang, J. Wu, C. Huang, L. Xu, "Attention-based multi-scale CNN-LSTM networks for lipreading," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 30, no. 12, pp. 4664–4675, 2020, doi:10.1109/TCSVT.2020.3017164.

[53] T. Broad, R. Vasudevan, and A. Chaturvedi, "Visual speech recognition for assistive communication devices," *Pattern Recognit. Lett.*, vol. 149, pp. 45–52, 2021, doi:10.1016/j.patrec.2021.01.013.